Core run I: free flight against the waypoint family

Run I tests a fair comparison of free flight and the reset waypoint family at equal family prior, together with the descent-climb family and the exact Davey reproduction. Every family has 8 seeds (7M particles per seed for free flight and waypoints, 3.5M for descent-climb and reproduction) and writes 100,000 hand-off rows per seed at 22:41 and 00:11 for end of flight. All inputs are staged at mh370-exchange/core/run-i/ (READY, 340 files, SHA-256 checked). 00:19 treatment: core, Davey both-BTO, no BFO.

1. Family result (equal prior over four families)

Family	ln Z	seed sd	ln BF	Share	Split-half	00:19 median
free flight	-135.78	0.35	-6.78	0.11 %	0.830	-36.83°
waypoints	-129.00	0.23	+0.00	99.82 %	0.916	-37.15°
descent-climb	-137.46	0.25	-8.46	0.02 %	0.794	-37.06°
reproduction	-136.73	0.18	-7.73	0.04 %	0.891	-36.73°

Mixture over the four families: 00:19 median −37.17°, 90 % interval −37.57° to −34.77°. No family reaches the 8-seed split-half floor of 0.924; waypoints is closest (0.916). The family log-evidences are stable to ±0.35 nats across seeds, so the ranking is robust; the shape of each latitude curve is not settled.

2. Why the waypoint family takes 99.8 %

Data	Gain vs free flight (nats)
Radar band VAMPI–MEKAR 18:15–18:21	+2.38
Radar points 18:04–18:22	+0.55
18:25:34 BTO (log-on); 18:28 BTO+BFO	-0.22
18:39 BFO (phone call)	+1.74
19:41 BTO+BFO	+1.29
20:41 BTO+BFO	-0.63
21:41–23:15	+0.76
00:11 BTO+BFO	+0.83
00:19 (two handshakes)	+0.08
Total	+6.78

The sequential log-evidence increments sum exactly to each family's ln Z, so the difference splits by data item (each increment is conditional on the earlier data). About a third of the lead (+2.38) comes from the radar band between VAMPI and MEKAR: the waypoint family flies that published segment exactly, and the band is scored across that line with sd 1.5 NM. The rest comes mainly from the 18:39 BFO (+1.74) and the 19:41 arc (+1.29): a turn at about 18:38 followed by a steady great circle fits them. Approximate sensitivity (dropping an item's increment): without the radar band the waypoint share is 0.981; also without 18:39, 0.90; also without 19:41, 0.73.

Turn timing is not forced. There is no time term in the prior. SAMAK, IGOGU and ANOKO lie 130–152 NM from MEKAR on the 94.42°E line, so at 450–510 kt the aircraft reaches them at 18:37–18:42; a fly-by starts about 2 minutes early. Earlier turns are in the prior (NILAM first at about 18:25; break-off at 20/40/60/80 % of the first leg at about 18:25, 18:29, 18:33, 18:37). The prior's turn times are discrete and most of its mass is at 18:37–18:42, so part of the 18:39 gain reflects where the fixes are. The paper should state this.

Run I: settings, waypoint plans, areas, traffic

3. Settings (Pete's rulings, 11–12 Oct 2026)

Speeds: free flight, waypoints and descent-climb use Mach 0.45–0.87 within 210–330 KCAS until 18:40, then cruise Mach 0.70–0.87 within 210–330 KCAS; physics limits take precedence. The reproduction family stays exactly Davey (Mach 0.73–0.84 throughout, no cruise CAS clip).

Waypoint family: VAMPI then MEKAR to 18:21, then a random route through the 26 approved 2014-epoch fixes (tiers A+B; tier C as sensitivity only). Fly-by and fly-over both sampled. Break-off 50/50 at 20/40/60/80 % on any leg; mean 3 legs, maximum 8. Equator: stop or continue 50/50; fixes allowed to 4.5°S. On the last leg the change of flight mode is 50/50 at the fix or uniform along the leg west of 96°E (no prior on when). After the change: Davey great-circle LNAV.

Common stack (all families): run D stack (run C sampler; fuel fixes s7 one-engine, s8 hold-then-taper, inop-flow fix; two tanks; Inmarsat ephemeris; radar in the likelihood; look-ahead hand-offs). Differences from Davey et al. (2016) are listed on the parameter page of each family report that follows.

4. Waypoint plans (posterior, 8 seeds)

Waypoint plan	Posterior share (seed sd)
SAMAK first	0.943 (0.023)
IGOGU first	0.041 (0.019)
NILAM first	0.012 (0.007)
ANOKO first	0.003 (0.001)
first leg fly-by	0.803 (0.060)
first-leg break-off at 40–80 %	0.189 (0.059)
last leg: change of mode part-way	0.521 (0.089)
two legs	0.759 (0.070)
three legs	0.149 (0.064)

5. Area of the position at 00:11 and 00:19 (sensitivity to the family)

Time	Set	Median lat	50 % area	90 % area	99 % area
00:11	all four (weighted)	-36.18°	7.0k km²	45k km²	109k km²
00:11	waypoints only	-36.18°	6.9k km²	45k km²	108k km²
00:11	free flight only	-35.68°	24.3k km²	73k km²	139k km²
00:19	all four (weighted)	-37.15°	7.5k km²	50k km²	119k km²
00:19	waypoints only	-37.15°	7.5k km²	50k km²	119k km²
00:19	free flight only	-36.67°	24.2k km²	77k km²	152k km²

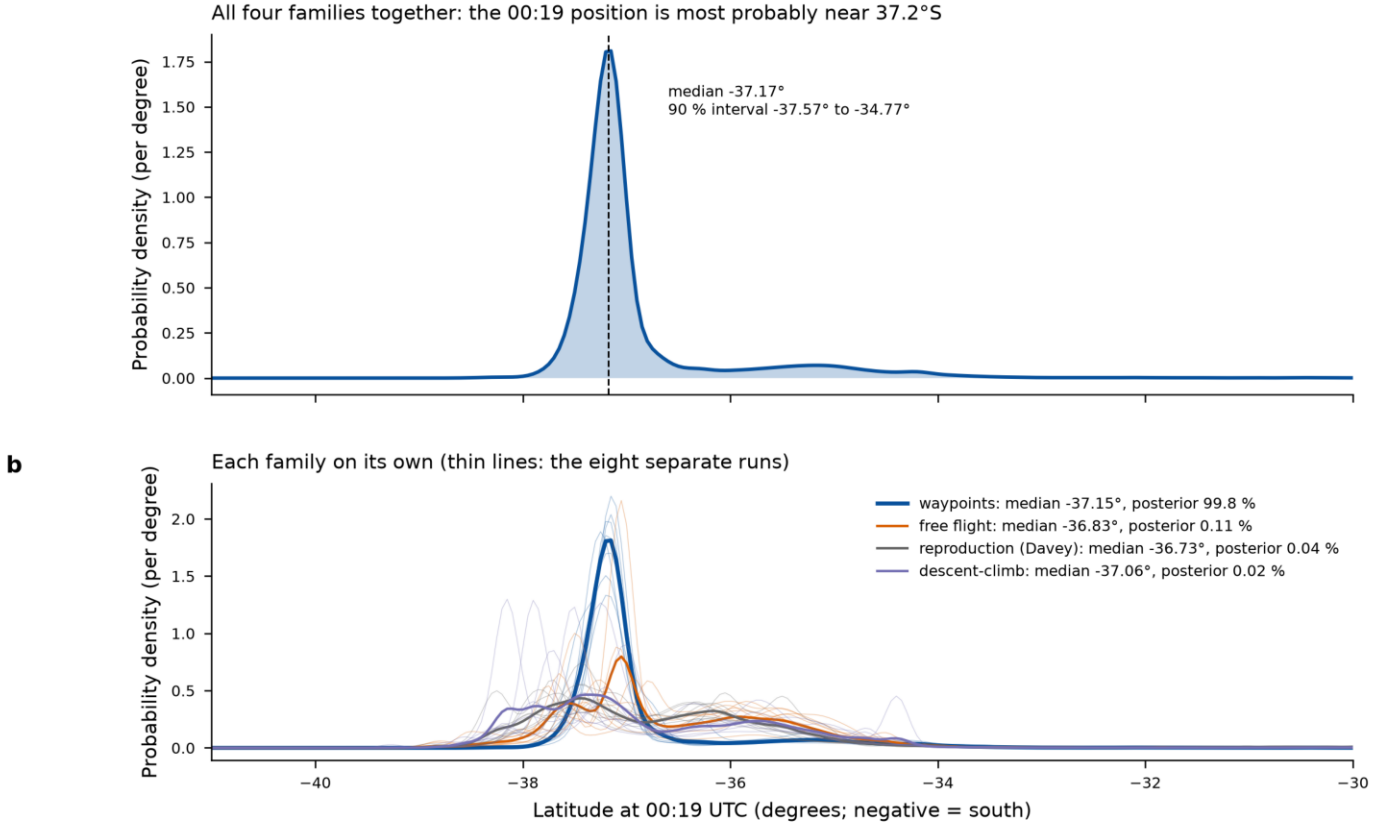
Run I: traffic and other checks

6. Separation from other traffic, 18:25–18:39

Family	Turn area	Traffic	Draws	Median (NM)	5 % (NM)	Close pass
free	7.2–7.55°N	IGO053	4598	10.2	3.7	1.0 %
free	7.2–7.55°N	UAE343	4598	11.9	5.2	0.1 %
free	north of 7.55°N	IGO053	8758	17.6	12.4	0.0 %
free	north of 7.55°N	UAE343	8758	26.5	9.8	0.0 %
free	south of 7.2°N	IGO053	2644	9.0	4.6	0.2 %
free	south of 7.2°N	UAE343	2644	20.7	12.0	0.0 %
waypoints	7.2–7.55°N	IGO053	1380	10.0	1.0	1.2 %
waypoints	7.2–7.55°N	UAE343	1380	16.3	8.1	0.0 %
waypoints	north of 7.55°N	IGO053	14396	15.2	14.9	0.0 %
waypoints	north of 7.55°N	UAE343	14396	30.6	25.6	0.0 %
waypoints	south of 7.2°N	IGO053	224	14.3	4.6	0.0 %
waypoints	south of 7.2°N	UAE343	224	16.6	11.7	0.0 %

Closest lateral approach of each posterior draw (2,000 per seed, 40 s trace). Close pass: within 5 NM laterally and 1,000 ft vertically. IGO053 positions are estimated (Langkawi SSR fit, back-propagated); UAE343 is ADS-C. Destination diagnostic (post-run only): no final great circle passes within 100 NM of NZFX; closeness to the South Pole is geometric. A particle refresh at 23:15 did not improve convergence (Δ ln Z −0.01, surviving lineages unchanged); more particles is the remedy for free flight.

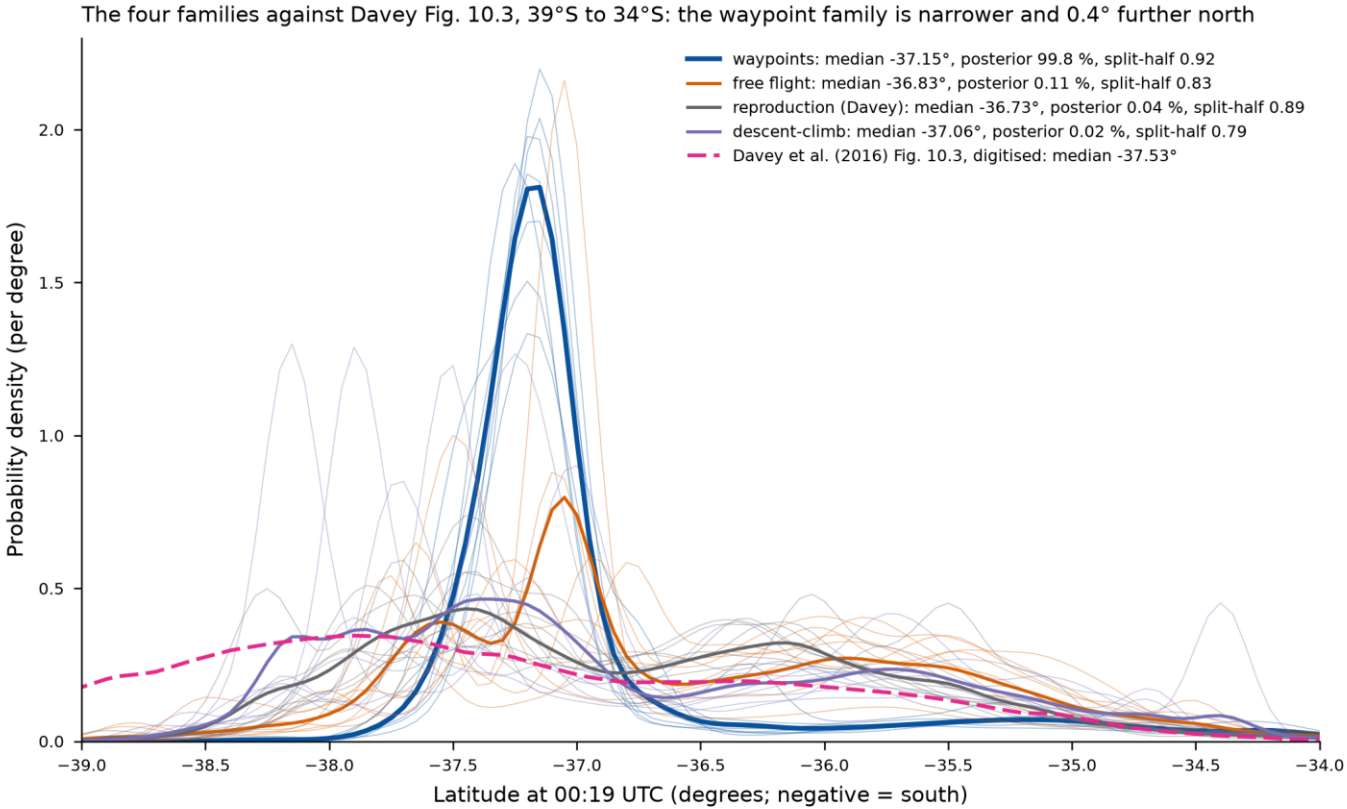
Figure A. 00:19 latitude: four families together, and each on its own



Simple version (ASD-STE100): The curves show where the aircraft probably was at 00:19 UTC, along the seventh satellite arc. Panel a combines the four groups of possible early flights. Each group has the same prior probability, and the data then give each group its share. The waypoint group has almost all of the probability, so panel a is almost the same as the waypoint curve in panel b. The thin lines are the eight separate runs. Where they do not agree, the result is less sure.

Technical version: Core run I, 12 Oct 2026; 8 seeds per family (7M particles per seed for free and waypoints, 3.5M for descent-climb and reproduction). Densities of 00:19 latitude from summary.json (0.1° smoothing), core 00:19 treatment: Davey both-BTO, no BFO. Mixture weights are family posteriors at equal prior from pooled log-evidence (waypoints −129.00, free −135.78, reproduction −136.73, descent-climb −137.46). Split-half over 35 balanced partitions: waypoints 0.916, free 0.830, reproduction 0.891, descent-climb 0.794; none reaches the 8-seed floor 0.924. Speed rule for free, waypoints and descent-climb: early Mach 0.45–0.87 within 210–330 KCAS to 18:40, cruise 0.70–0.87 within 210–330 KCAS; reproduction Davey Mach 0.73–0.84. One-engine phase (s7 + s8) provisional.

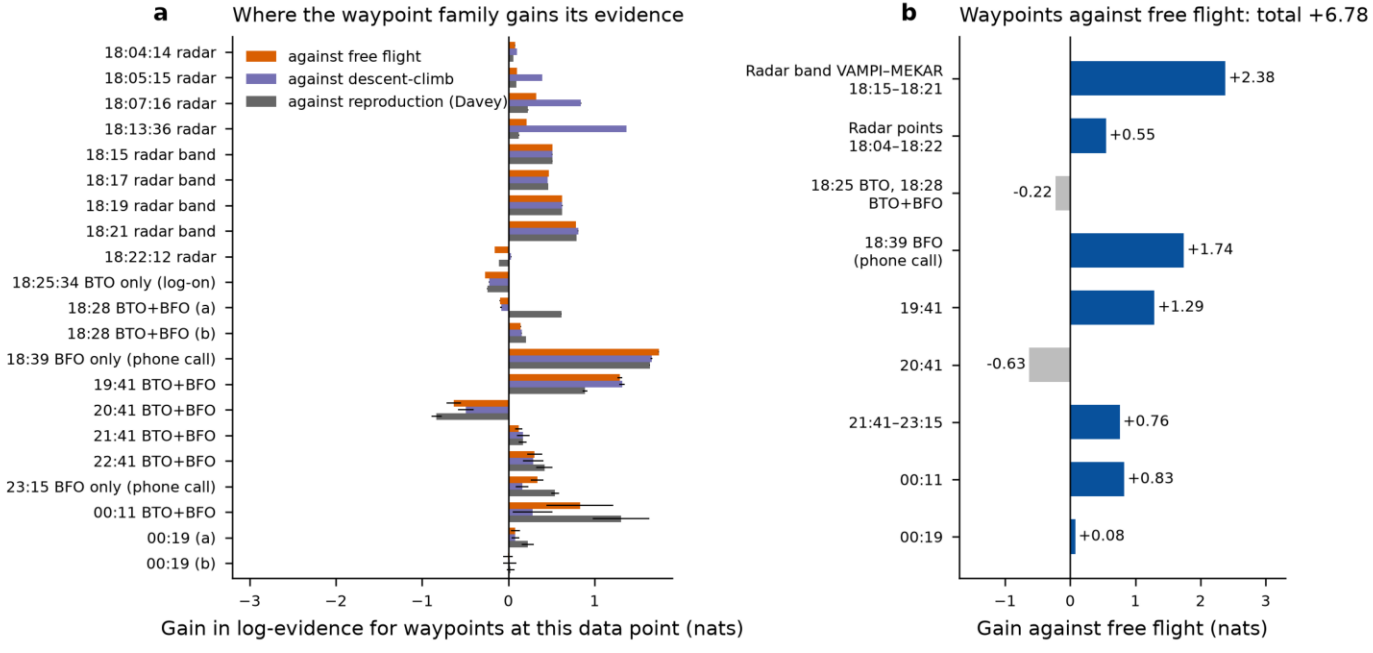
Figure B. 00:19 latitude, 39°S–34°S, with Davey Fig. 10.3



Simple version (ASD-STE100): The thick curves show where the aircraft probably was at 00:19 UTC for each group of possible early flights. The thin curves are the eight separate runs of each group. When the thin curves of a group are close together, the result for that group is sure. The waypoint group has 99.8 % of the probability. The dashed curve is the published result of Davey and others (2016, Figure 10.3).

Technical version: Core run I, 12 Oct 2026; 8 seeds per family (7M particles per seed for free and waypoints, 3.5M for descent-climb and reproduction). Densities of 00:19 latitude from summary.json (0.1° smoothing); core 00:19 treatment, Davey both-BTO, no BFO. Family posteriors at equal prior from pooled log-evidence. Split-half is the mean overlap over 35 balanced seed partitions; the 8-seed floor is 0.924. Same data as Figure run-i-0019-latitude panel b, with the latitude range 39°S–34°S and the vertical scale doubled. Davey et al. (2016) Fig. 10.3 as digitised in the engine's reference (summary.json), on the same latitude grid; its southern tail continues past 39°S (2.5 % quantile 39.38°S).

Figure C. Where the waypoint family gains its evidence

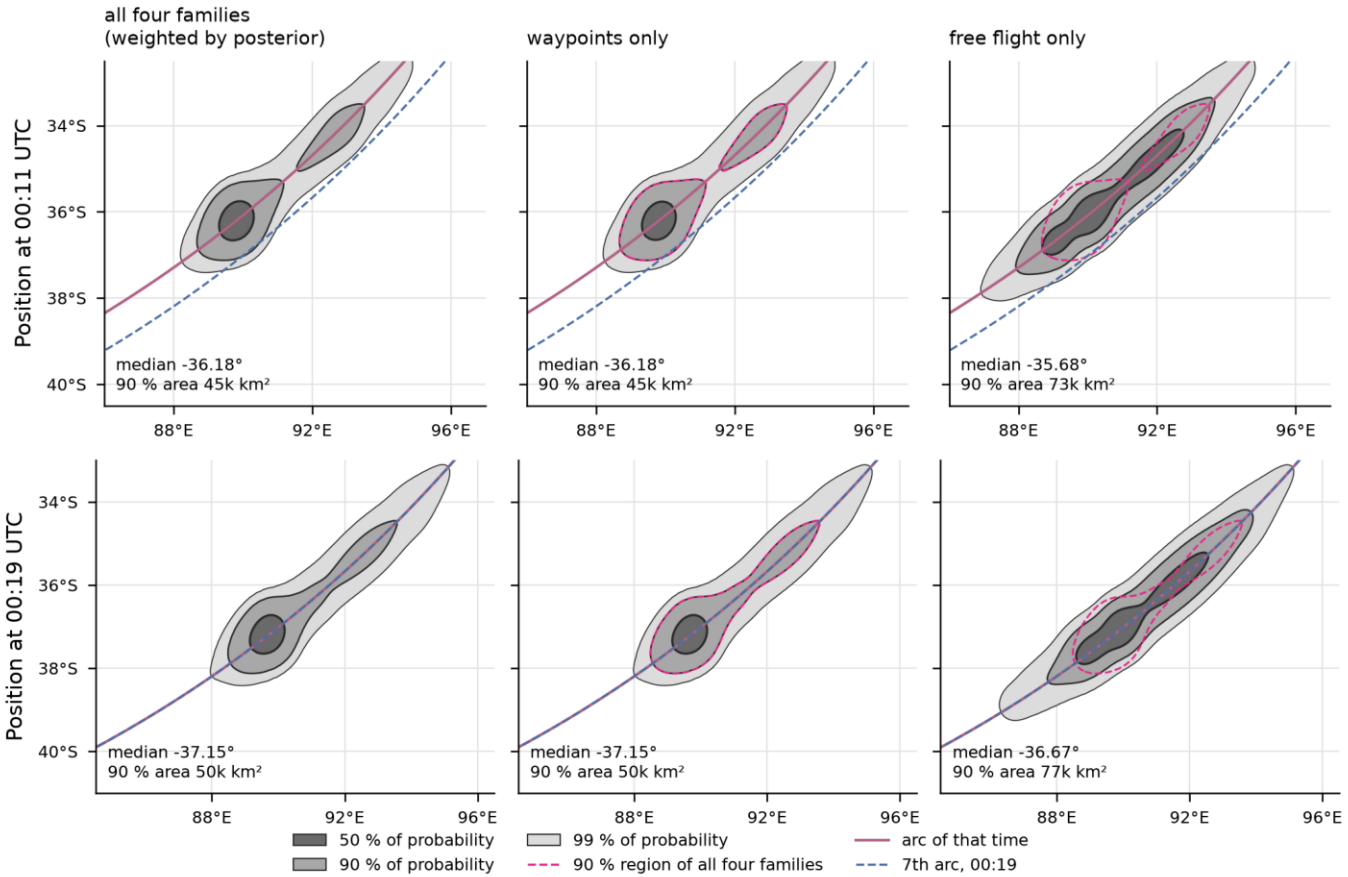


Simple version (ASD-STE100): Each bar shows how much better the waypoint paths agree with one item of data than the other paths do. A bar to the right helps the waypoint paths. The largest gains are at the radar band between VAMPI and MEKAR (18:15 to 18:21), at the frequency data of the 18:39 phone call, and at the satellite data at 19:41. The waypoint paths fly exactly on the line from VAMPI to MEKAR, and the radar band is on that line. The total gain against free flight is 6.8. This gives the waypoint paths 99.8 % of the probability when the four groups have the same prior probability.

Technical version: Core run I, 8 seeds per family. Per data point, the stratum's sequential log-evidence increment ln p(y_k | y_1..k-1), modes combined by their prior weights; bars are waypoints minus the other family, mean over seeds, error bars the seed standard deviation (zero where all seeds share the same early particles). The increments sum exactly to the difference in total log-evidence; each is conditional on all earlier items, so the split depends on data order. Radar band = across-track scores against the VAMPI–MEKAR great circle at four times, sd 1.5 NM (radar-full.toml). The waypoint family flies the VAMPI–MEKAR prefix with no lateral randomness from 18:01:49 to 18:21; the other families carry lateral dispersion there. 18:39 and 23:15 are C-channel calls: BFO only, no BTO. 00:19: core, Davey both-BTO no-BFO.

Figure D. Area of the position at 00:11 and 00:19: all families, waypoints only, free flight only

Where the aircraft was at 00:11 and 00:19: waypoint paths give a smaller area than free flight

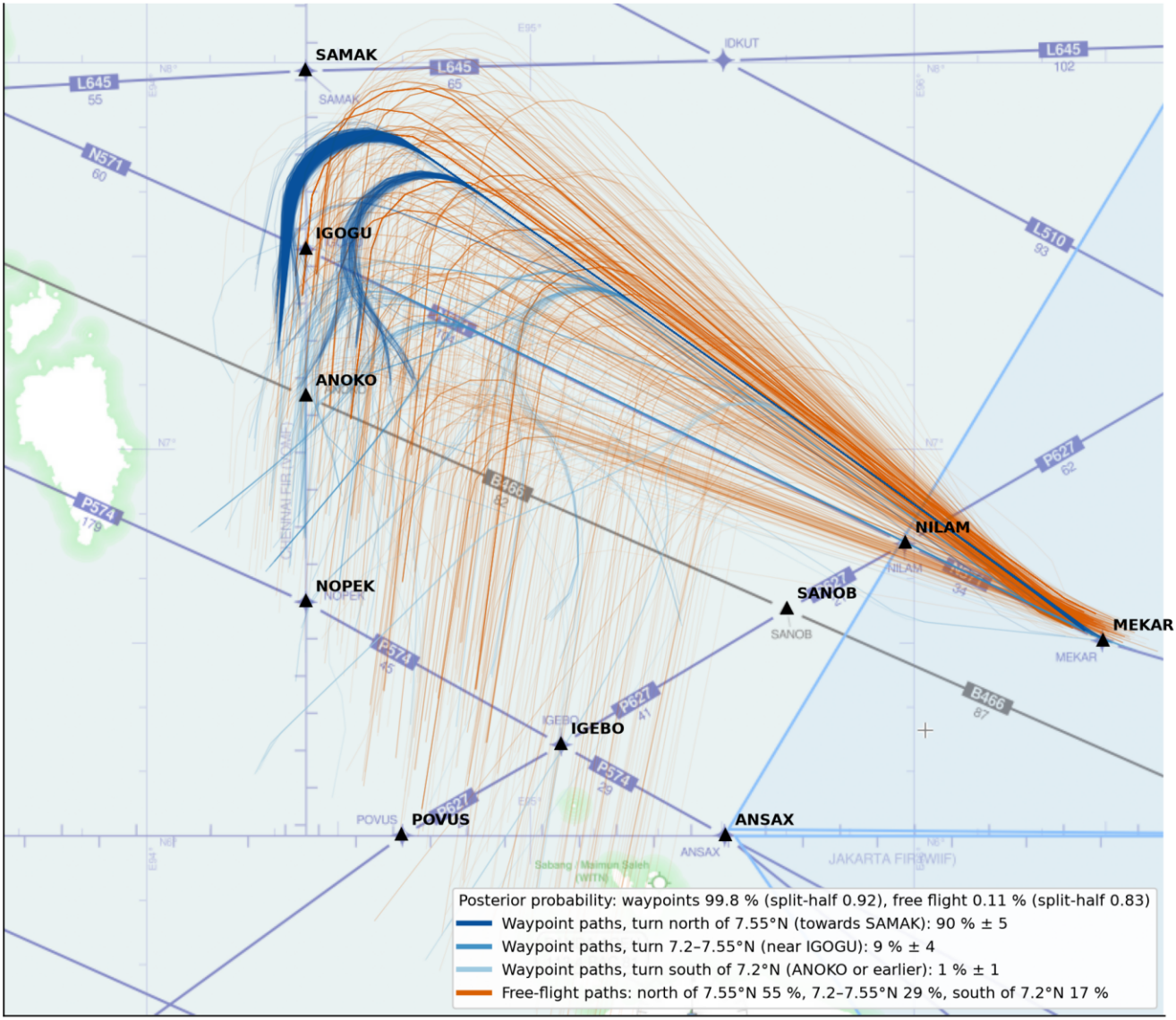


Simple version (ASD-STE100): The grey areas show where the aircraft probably was at 00:11 UTC (top) and 00:19 UTC (bottom). Darker grey has more probability. The left column uses all four groups of early flights, each with its share of probability from the data. The middle column uses only the waypoint paths. The right column uses only the free paths. The dashed pink line is the 90 % area of the left column, for comparison.

Technical version: Core run I, 8 seeds per family. 00:11: positions interpolated from the 2,000 posterior route draws per seed (smoothing distribution); 00:19: final.npy states, thinned to 1.5M rows per seed, with their weights. Family weights at equal prior from pooled log-evidence (waypoints 0.9982, free 0.0011, reproduction 0.0004, descent-climb 0.0002); seeds equally weighted within a family. 0.1° grid, 0.3° Gaussian smoothing, highest-density regions at 50/90/99 % (report/epoch_map.py convention). Areas on the smoothed grid. 00:19 treatment: core, Davey both-BTO, no BFO. One-engine phase provisional.

Figure E. Posterior paths 18:22–18:42 on the en-route chart

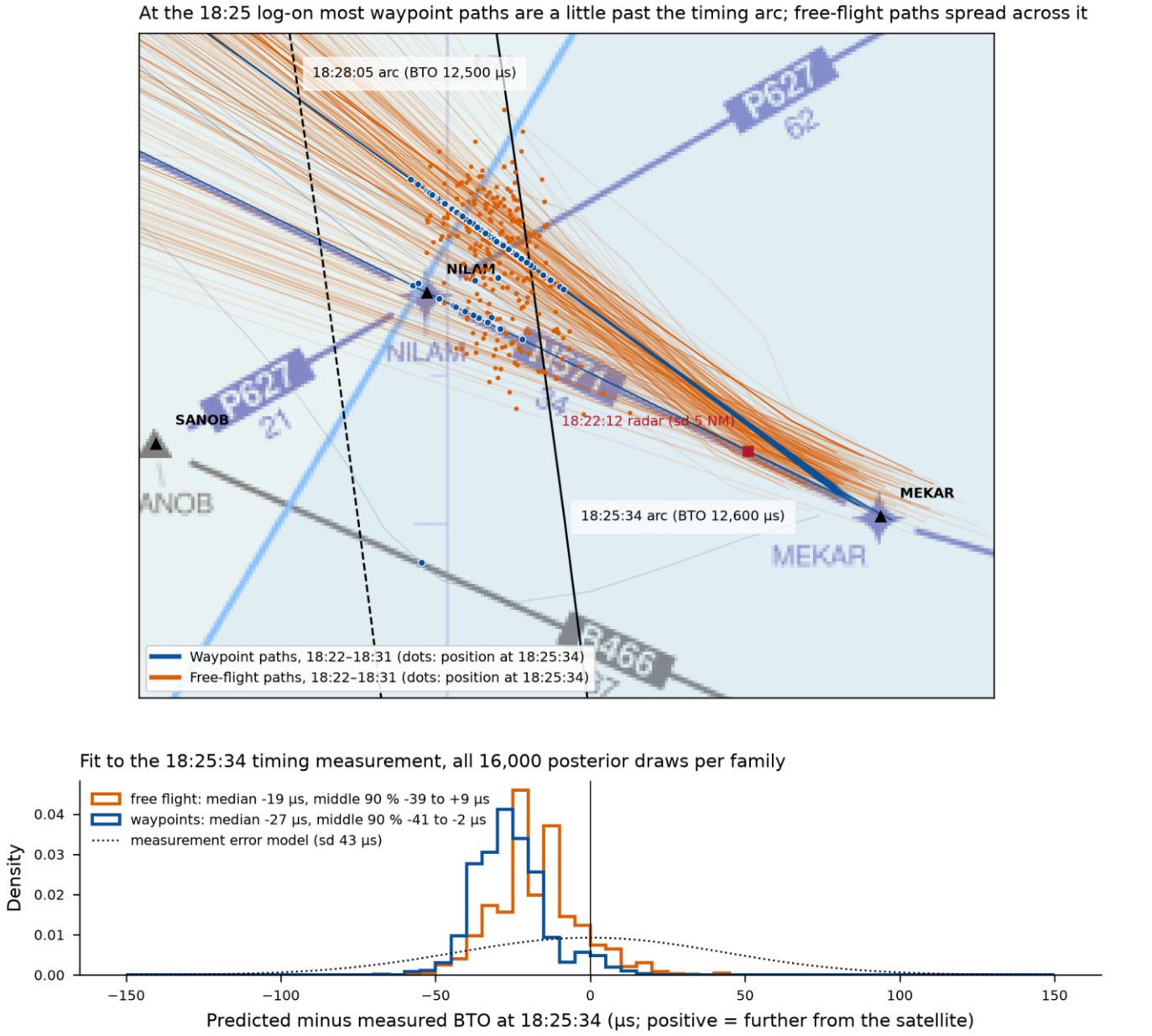
Paths that follow waypoints, which the data favour, turn south between IGOGU and SAMAK at 18:37–18:39 UTC



Simple version (ASD-STE100): The lines show possible flight paths from 18:22 to 18:42 UTC. Each line is one sample from the result. Blue lines are paths that fly to fixed waypoints. Orange lines are paths that fly freely. We show the same number of lines for the two groups, but the data give 99.8 % of the probability to the waypoint paths. The percentages show how many paths of each group turn in each area. The ± value is the change between the eight separate runs. The chart is for orientation only.

Technical version: Core run I, 12 Oct 2026; equal prior over four families (free flight, waypoints, descent-climb, Davey reproduction); family posterior from pooled log-evidence (waypoints −129.00, free −135.78). 8 seeds × 7M particles; 2,000 posterior route draws per seed, 40 s trace 18:22:00–18:42:40; 150 draws per seed per family drawn. Turn area classified by the northernmost latitude reached (≥7.55°N, 7.2–7.55°N, <7.2°N); ± is the seed standard deviation. Split-half = mean overlap over all 35 balanced seed partitions of the 00:19 latitude density (8-seed floor 0.924). Speed rule: early phase Mach 0.45–0.87 within 210–330 KCAS to 18:40, cruise 0.70–0.87 within 210–330 KCAS. 26 fixes (2014-epoch tiers A+B); fly-by and fly-over sampled; break-off 50/50 at 20/40/60/80 % on any leg. 00:19 treatment: core, Davey both-BTO no-BFO. Background: chart supplied by Pete (current cycle), affine fit to 12 fixes, residual ≤1.4 px; fix symbols at the positions used in the runs.

Figure F. Paths around the 18:25:34 log-on, MEKAR to NILAM



Simple version (ASD-STE100): The top map shows possible flight paths from 18:22 to 18:31 UTC near MEKAR and NILAM. The dots show where each path is at 18:25:34, when the satellite system logged on again. The black lines show where the timing measurements put the aircraft. The bottom chart shows the difference between each path and the 18:25:34 measurement. The waypoint paths agree a little less well at 18:25:34 than the free paths, because they are all on one line and most of them are a little too far west at that time. This costs the waypoint paths 0.3 of evidence. They gain much more later.

Technical version: Core run I, 8 seeds × 2,000 posterior route draws per family; 40 s trace from 18:22:00, linearly interpolated to 18:25:34; 60 draws per seed shown on the map. BTO predicted with report/satcom_model.py (Inmarsat ephemeris, terminal altitude from the trace) against the logged 12,600 µs; sd 43 µs at 18:25:34 (log-on, BFO excluded), 29 µs at 18:28. Arcs drawn from the engine's reference_arcs (run.json). Evidence increments at 18:25:34 (waypoints minus free) −0.27 nats; 18:22:12 radar −0.16; 18:28 −0.10 and +0.14. Chart background supplied by Pete, affine fit to 12 fixes.

Where was MH370 at 00:19 UTC?

A reproduction of the DSTG Bayesian aircraft-position pdf (Davey et al. 2016, Fig. 10.3)

Core run I, family 'waypoints'. In the early flight, the aircraft flies VAMPI and MEKAR, then a random route through 26 published waypoints of 2014. It can turn before a waypoint or fly over it, and it can leave a leg early. Speeds are as for free flight. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight.

This report recreates the probability density function (pdf) of the aircraft's latitude at the final satellite handshake, 00:19 UTC on 8 March 2014, from the model published by Davey, Gordon, Holland, Rutten and Williams, Bayesian Methods in the Search for MH370 (Springer, 2016). The aircraft state is filtered from the penultimate radar point at 18:01:49 UTC through all Inmarsat burst timing (BTO) and burst frequency (BFO) measurements, using the published cruise, manoeuvre and measurement models.

With BTO and BFO the recreated pdf has median 37.15°S and mode 37.15°S, with 95% of probability between 34.05°S and 37.63°S. The published curve has median 37.53°S, mode 37.90°S and 95% range 34.90–39.38°S. The two densities overlap by 39%. This curve is centred 0.4° further north and carries 11% of probability in the published northern shoulder between 34.5°S and 36.5°S (25% in the book).

Relative to the same family in run D, the median moves 0.07° north, the 34.5–36.5°S shoulder holds 11% of probability (base 15%), and the two curves overlap by 79%.

The curve pools 8 independent replicates of 7,000,000 particles. Single replicates overlap each other by 75%–93% and their medians span 0.13°. Over all balanced partitions of the 8 replicates the split-half overlap averages only 91.6% (range 84.5% to 94.8%) against the 8-replicate floor of 0.924, so the pooled curve is NOT converged at this particle count and quantities read off it carry that caveat.

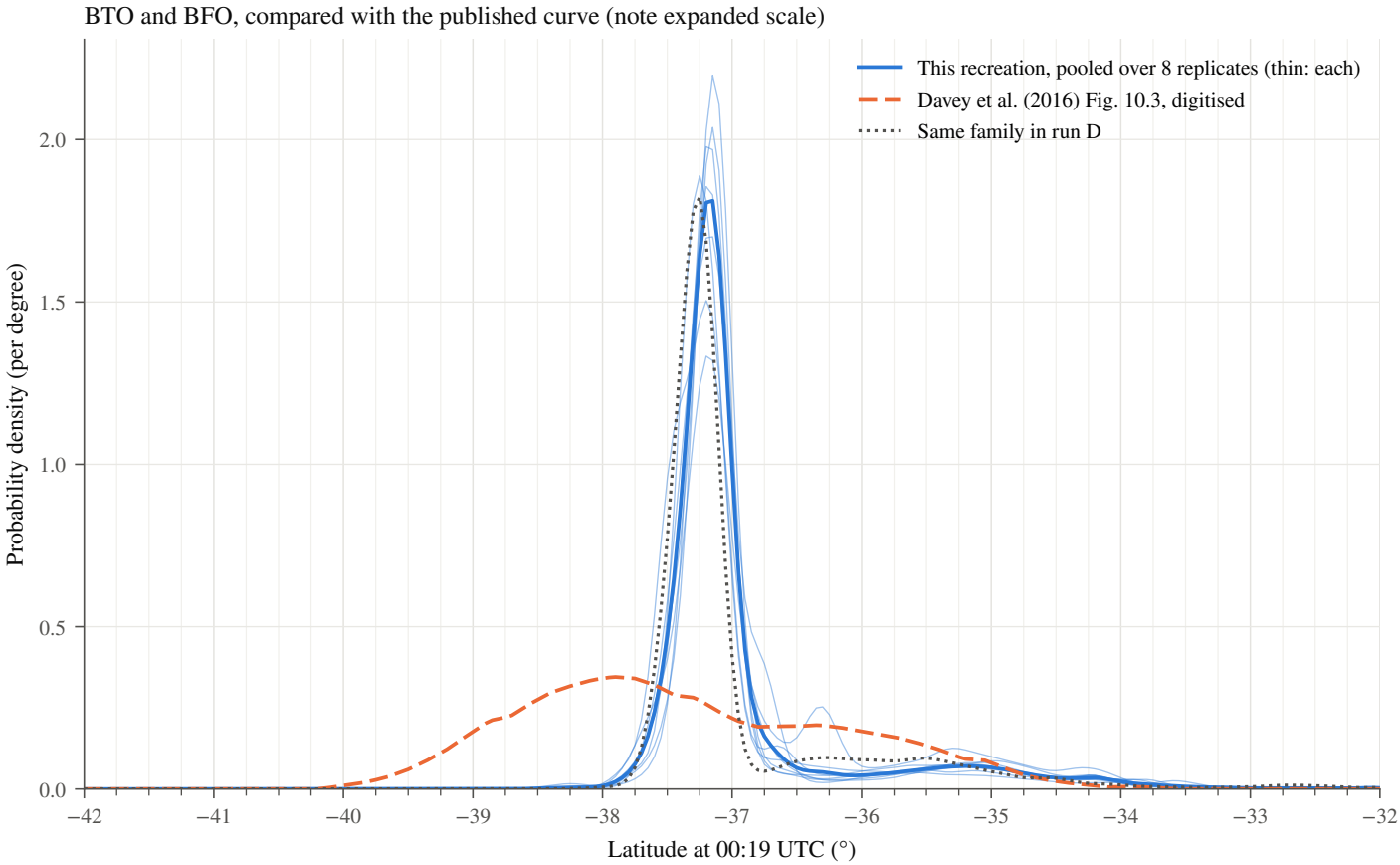


Figure 1. Latitude pdf of the aircraft at 00:19:37 UTC, the form of Davey et al. Fig. 10.3. BTO and BFO, with the published curve. Densities are weighted particle histograms smoothed with a 0.1° Gaussian; each particle carries its own altitude. Evidence: BTO at every epoch except the 18:39 and 23:15 C-channel calls; BFO (bottom only) at 18:28, 18:39, 19:41–22:41, 23:15 and 00:11. This run also uses: primary-radar positions and track segments 18:04-18:22 scored as measurements; fuel model internal-v1, two tanks, one-engine flight after the first flame-out (hold-taper drift-down), paths dry before 00:11 rejected. No debris, drift, search or end-of-flight evidence is used. Run conditions as stated at the top of this page.

Plain: Core run I, family 'waypoints'. In the early flight, the aircraft flies VAMPI and MEKAR, then a random route through 26 published waypoints of 2014. It can turn before a waypoint or fly over it, and it can leave a leg early. Speeds are as for free flight. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.916, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.
Technical: run next-i-waypoints; stack: run D stack + speed rule (see the method page); binary ee34559, x86_64-linux (deskstar); seeds 1-8 x 7,000,000 particles; ln Z -129.00; split-half mean over 35 balanced partitions 0.916 (range 0.845-0.948) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.

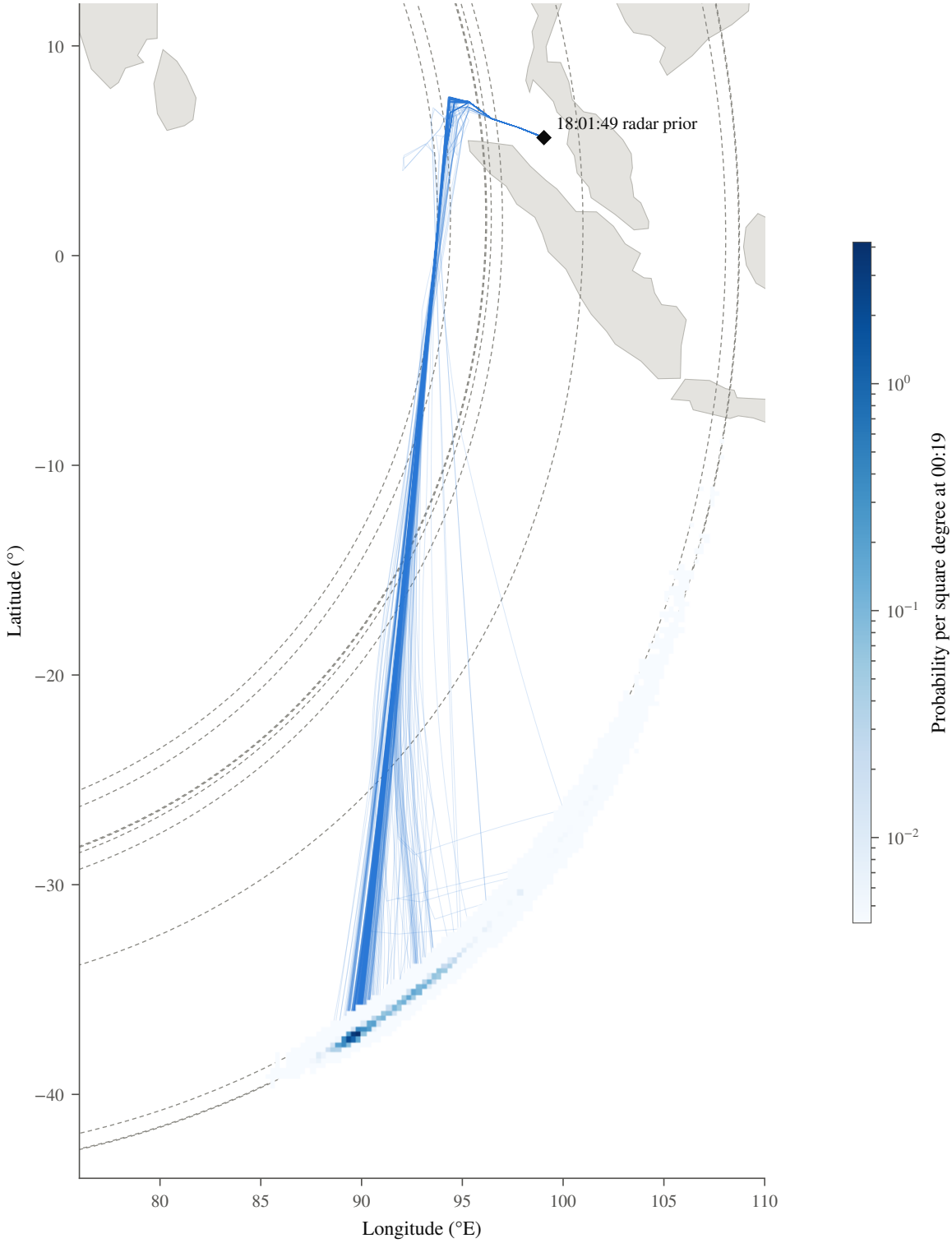
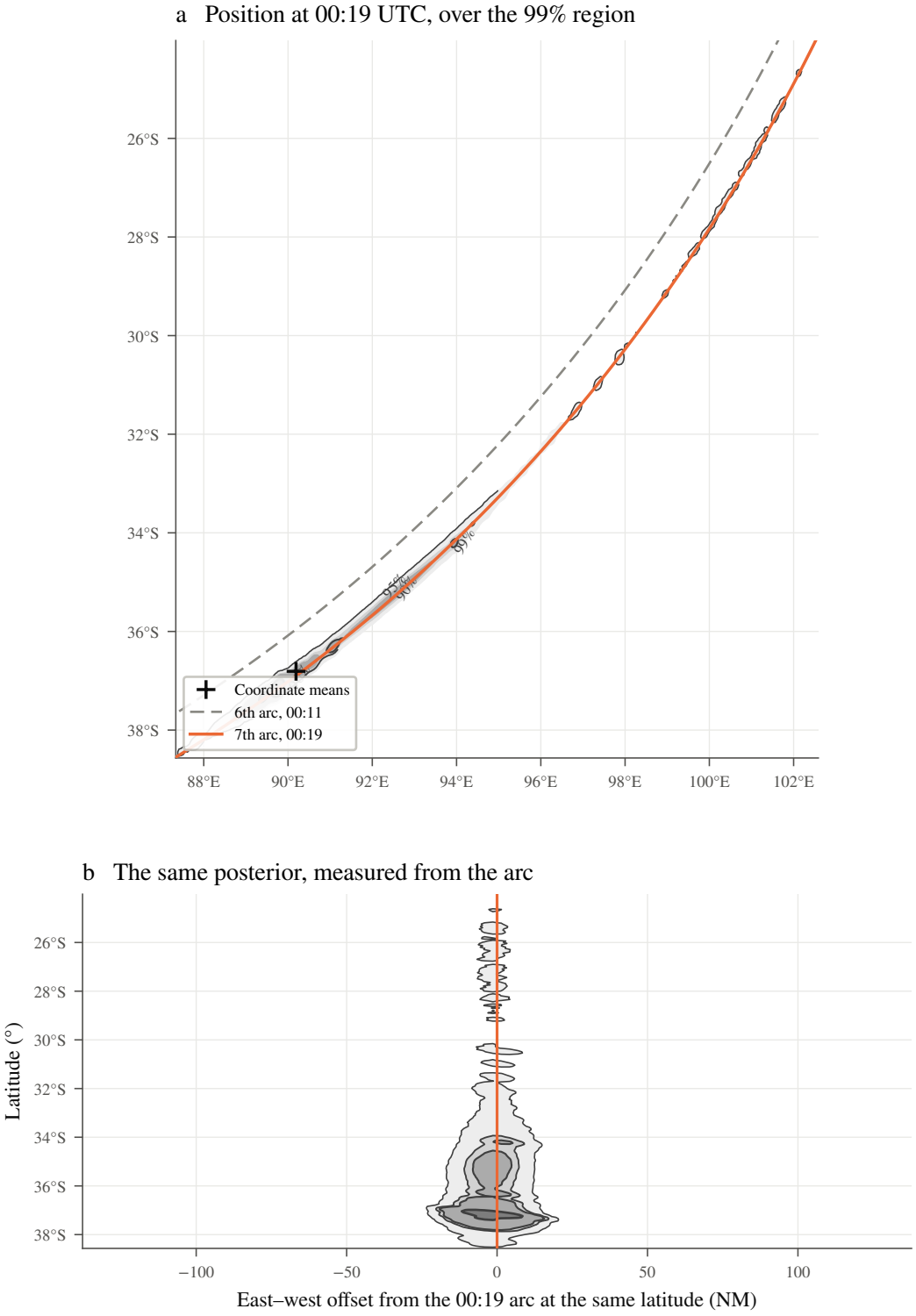


Figure 2. Posterior aircraft position at 00:19 UTC with BTO and BFO, pooled over 8 replicates (blue shading, logarithmic), and 400 routes drawn in proportion to posterior weight from replicate seed 1 (thin lines, 10-minute vertices). Dashed grey curves are the BTO arcs at a reference altitude of 35,000 ft; they are drawn for orientation only. Land: Natural Earth 1:110m.

Plain: Core run I, family 'waypoints'. In the early flight, the aircraft flies VAMPI and MEKAR, then a random route through 26 published waypoints of 2014. It can turn before a waypoint or fly over it, and it can leave a leg early. Speeds are as for free flight. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.916, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.
Technical: run next-i-waypoints; stack: run D stack + speed rule (see the method page); binary ee34559, x86_64-linux (deskstar); seeds 1-8 x 7,000,000 particles; ln Z -129.00; split-half mean over 35 balanced partitions 0.916 (range 0.845-0.948) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.



DIFFERENCES FROM DAVEY

Mach after an acceleration, uniform on: [0.7, 0.87]

Manoeuvre integration step: 5

Position sd: 0.5

Initial Mach set point: -

Prior mean position and track:

BTO noise sd, R600: 63 at 1 epoch

Vertical rate in the BFO model: True

Weather source: era5-wind-temperature.bin

Magnetic declination source: igrf14-declination.bin

Resampling scheme: one filter per autopilot …

Cost-index speed mode:

Fuel and endurance: {'factor_mean': 1.0004, '…

Fuel proposal: endurance (deadline unix …

book values: see the parameter page

POSTERIOR AT 00:19

median 37.15°S, mode 37.15°S

95% 34.05–37.63°S

replicates 8, split-half 0.869

largest single draw 0.00% of weight

99% region 13,082 NM²

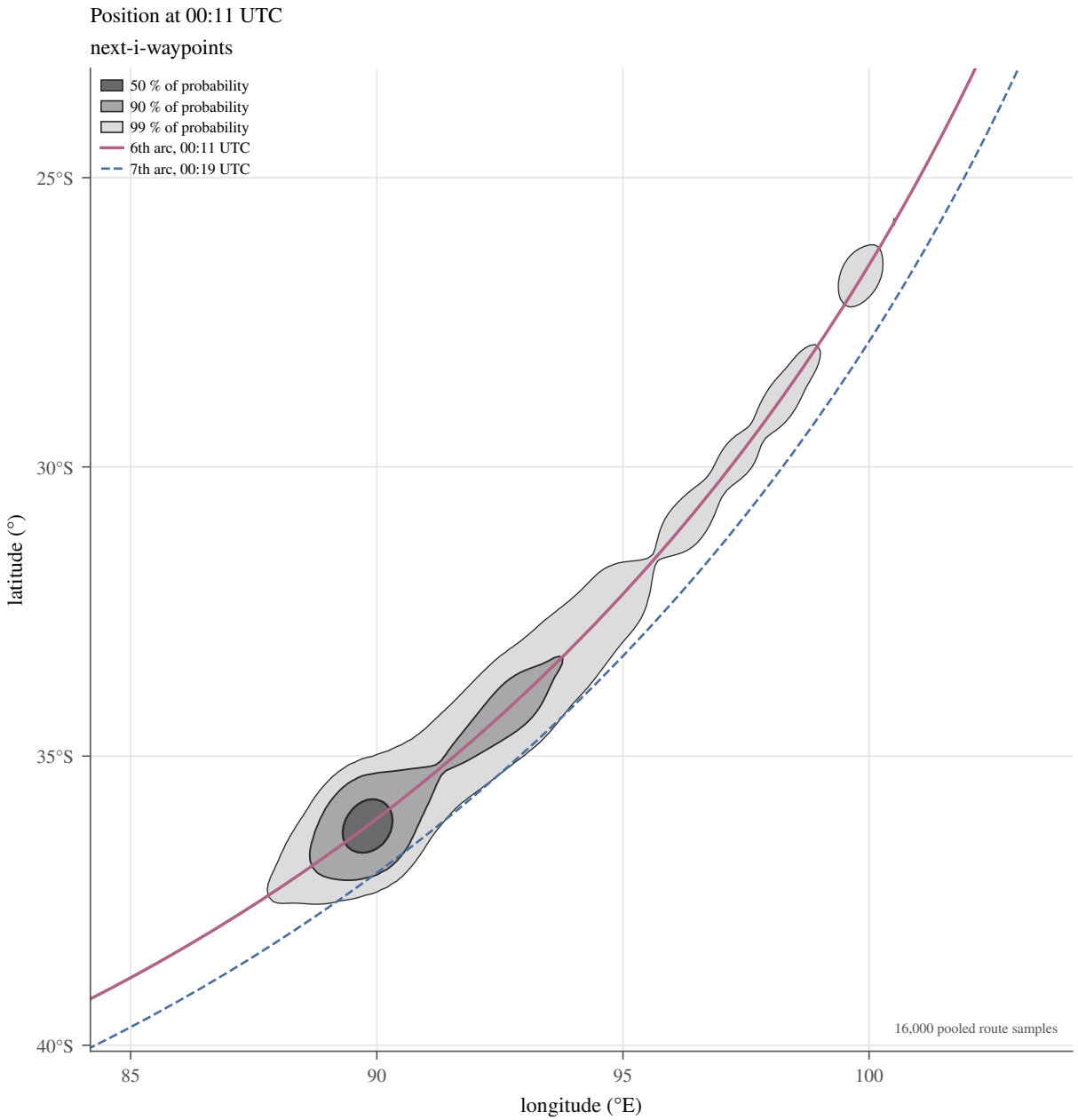
95% region 5,276 NM²

90% region 3,376 NM²

50% region 315 NM²

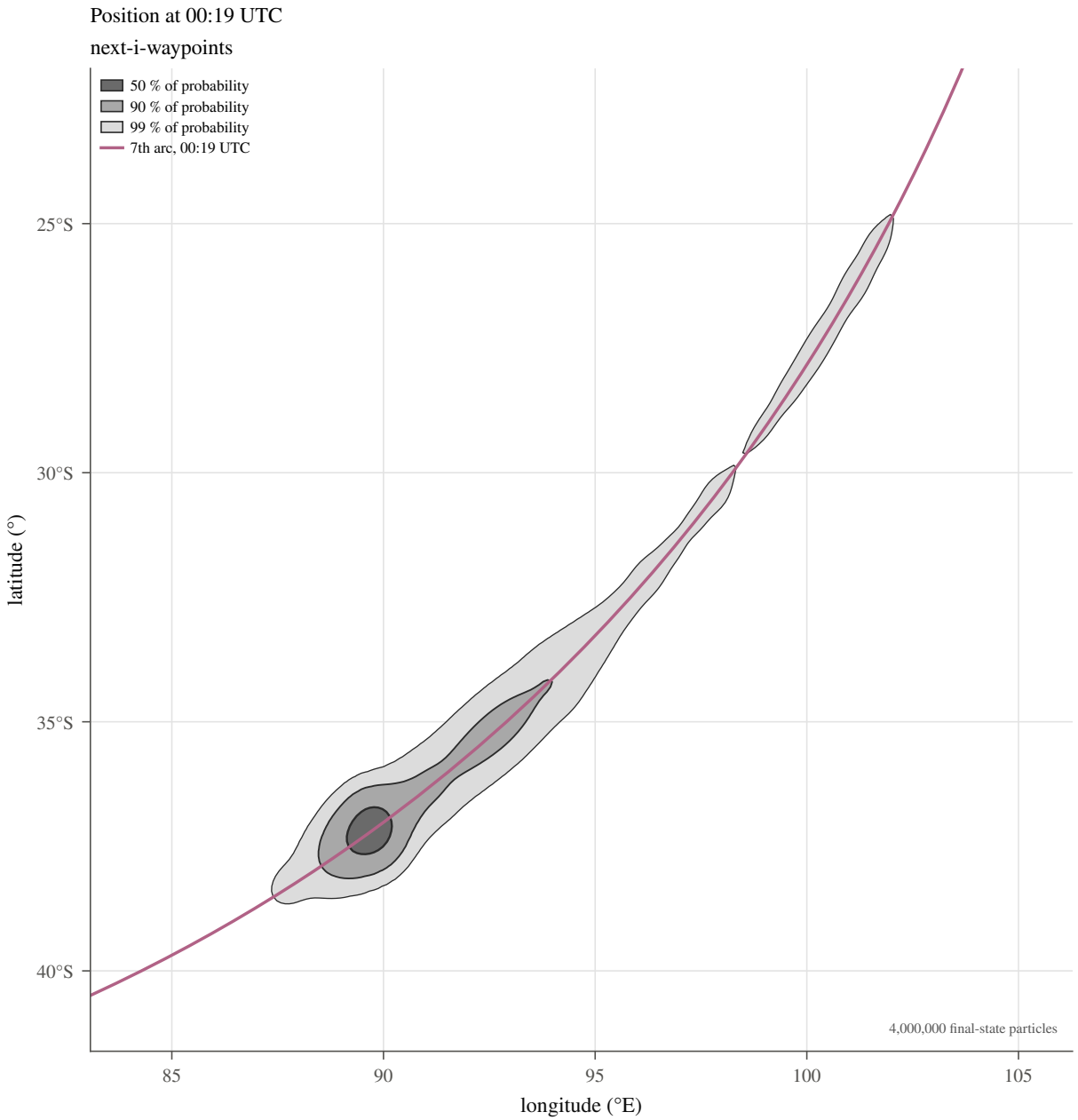
Figure 3. Joint latitude-longitude posterior at 00:19 UTC, drawn as highest-posterior-density regions: the 50% curve bounds the smallest area holding 50% of the probability, and so on outward, so a region here is the smallest place to look for that confidence rather than an interval in either coordinate alone. Binned at 0.05° and smoothed over three bins. Panel a spans the bounding box of the 99% region, so sparse outlying cells are excluded; enclosed areas (NM², at the view's mean latitude) are 99% 13,082, 95% 5,276, 90% 3,376, 50% 315. Panel b replots the same particles against their east-west offset from the 00:19 arc, because the region is about 0.3° wide and 15° long and the inner contours are not separable at a true aspect ratio. Arcs are the reference BTO loci at 35,000 ft; 00:19 is the 7th. Land: Natural Earth 1:110m.

Plain: Core run I, family 'waypoints'. In the early flight, the aircraft flies VAMPI and MEKAR, then a random route through 26 published waypoints of 2014. It can turn before a waypoint or fly over it, and it can leave a leg early. Speeds are as for free flight. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.916, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.
Technical: run next-i-waypoints; stack: run D stack + speed rule (see the method page); binary ee34559, x86_64-linux (deskstar); seeds 1-8 x 7,000,000 particles; ln Z -129.00; split-half mean over 35 balanced partitions 0.916 (range 0.845-0.948) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.



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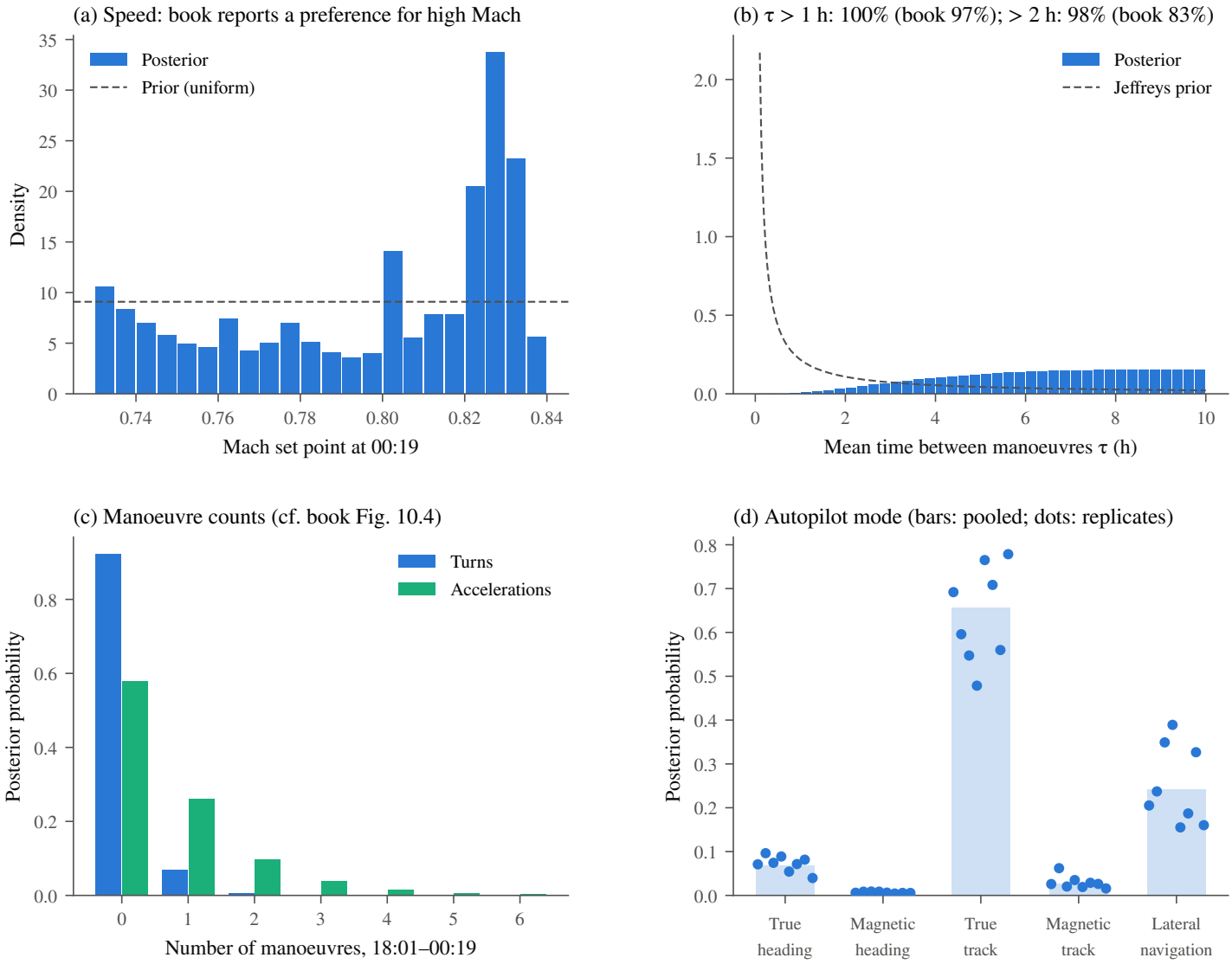
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Posterior behaviour compared with the book



	This recreation	Davey et al. (2016)
Median latitude	37.15°S	37.53°S
Mode	37.15°S	37.90°S
95% interval	34.05–37.63°S	34.90–39.38°S
95% width	3.58°	4.47°
Overlap with published pdf	39% (replicates 34%–50%)	—
Median Mach at 00:19	0.809	upper part of 0.73–0.84

Figure 3 and Table 1. Posterior summaries with BTO and BFO, pooled over replicates, alongside the statements in Davey et al. (2016), sec. 10.2–10.4. Book values of the latitude pdf come from the digitised Fig. 10.3 (bottom); other book values are quoted from the text.

Plain: Core run I, family 'waypoints'. In the early flight, the aircraft flies VAMPI and MEKAR, then a random route through 26 published waypoints of 2014. It can turn before a waypoint or fly over it, and it can leave a leg early. Speeds are as for free flight. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.916, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.
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Method, convergence and reproduction

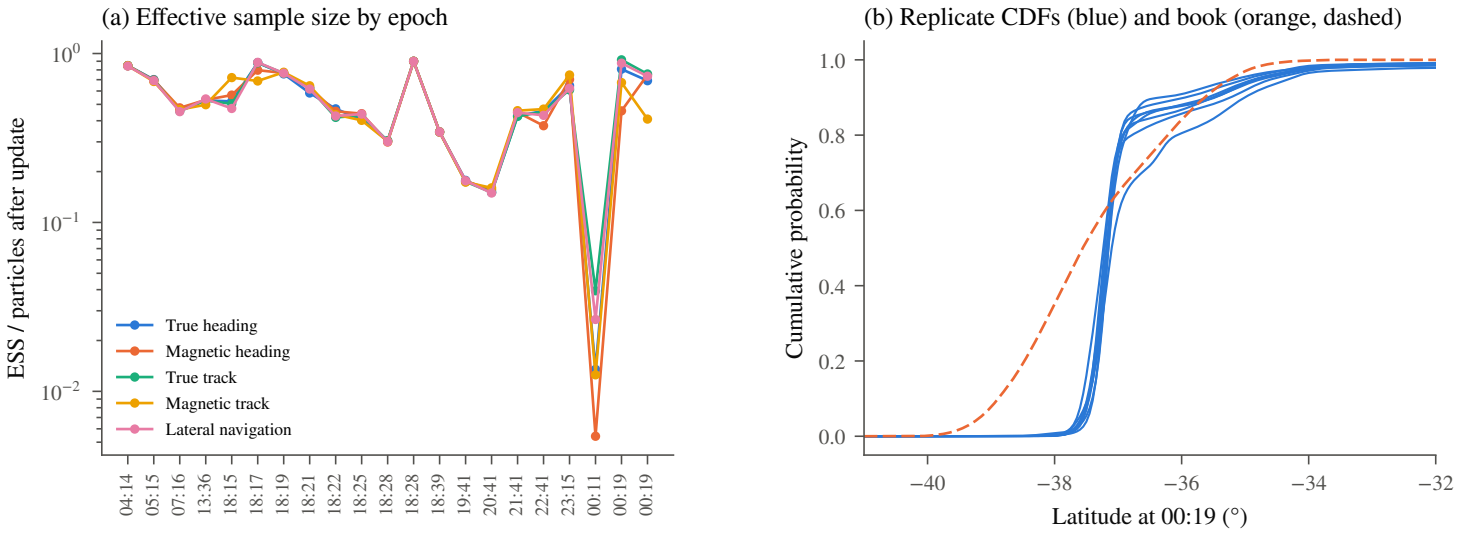


Figure 4. (a) Effective sample size after each measurement update, averaged over replicates, per autopilot-mode filter. The narrowest bottlenecks are the 18:39 C-channel BFO (the southward turn) and the 19:41 BTO arc. (b) Latitude CDFs of the 8 independent replicates; the largest difference between any two is 0.196 (Kolmogorov distance).

Model. The state, dynamics and likelihood follow Davey et al. (2016) chapters 4–8: prior at 18:01:49 UTC with 0.5 NM position and 1° track uncertainty, Mach uniform on 0.73–0.84, altitude on 1,000 ft levels from 25,000 to 43,000 ft; five autopilot modes (true/magnetic heading, true/magnetic track, lateral navigation with a one-off reversion to heading hold); Ornstein–Uhlenbeck fluctuations of Mach, control angle and wind error with the published rates; turns, speed changes and climbs arriving with exponential gaps of common mean τ (Jeffreys prior on 0.1–10 h) at 15° bank, 0.1 Mach per minute and 4,000 ft per minute. BTO is Gaussian with the tabulated per-message errors (29, 43 or 63 µs); BFO has 7 Hz noise and an unknown constant bias (prior 150 ± 25 Hz) marginalised exactly per particle with a Kalman filter.

Substitutions. The book's ACCESS-G weather is replaced by ERA5 winds and temperatures; magnetic declination is IGRF-14. The prior mean position and track are reconstructed from the book's radar figures because they are not tabulated. Davey et al. used variable-rate branching; this recreation uses fixed-size particle filters with systematic resampling when ESS falls below 50% of the particles. The book warns that conventional resampling collapses static parameters (sec. 10.4). Both are therefore handled exactly: one filter per autopilot mode, combined in proportion to its estimated marginal likelihood, and a Gibbs refresh of τ from its path conditional after every resampling step. Without them, a plain bootstrap filter of one million particles retained only about one hundred distinct prior draws, and an earlier recreation in this project found tails and turn counts that changed with the random seed.

Run. Configuration next-i-waypoints; particles per mode 1,000,000, 500,000, 2,500,000, 500,000, 2,500,000 (order as in Figure 3d), 7,000,000 per replicate; BTO+BFO seeds 1, 2, 3, 4, 5, 6, 7, 8; code revision ee34559; 88 threads; total runtime 186.5 min; peak memory 35,188 MiB. Hypotheses enabled: radar-fix (parameters in run.json). Sensitivity run 'next-i-waypoints', varying cas_envelope_kt (not declared) to [210.0, 330.0]; route_sampler (not declared) to {'break_off_fractions': [0.2, 0.4, 0.6, 0.8], 'break_off_...; routes {'first': [['VAMPI', 'MEKAR', 'NILAM', 'IGOGU'], ['VAMPI'... to (not declared); mach_range [0.73, 0.84] to [0.7, 0.87]; particles_per_mode [500000, 250000, 1250000, 250000, 1250000] to [1000000, 500000, 2500000, 500000, 2500000]; replicates 4 to 8.

Reproduce with one command from the repository root: make hypothesis H=radar-fix

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Parameters and model choices against Davey et al. (2016)

43 parameters and choices. 30 match the published model; 13 differ and are marked. A difference is not by itself a defect: some are deliberate and documented (ERA5 for the unavailable ACCESS-G, IGRF-14 for NOAA, the sampler), some are the subject of the run (a widened Mach range), and some are worth fixing. Page numbers refer to the book; the full table with citations and notes is written beside this report as parameters.csv.

	Parameter	This run	Davey et al. (2016)
	Dynamics		
	Mach OU reversion rate (1/s)	0.01058	0.01058
	Mach OU noise strength (1/s)	2.05e-07	2.05e-07
	Control-angle OU reversion rate (1/s)	0.009792	0.009792
	Control-angle OU noise strength (rad^2/s)	4.074e-08	4.074e-08
	Wind-error OU reversion rate (1/s)	0.001087	0.001087
	Wind-error OU noise strength (kn^2/s)	0.07021	0.07021
	Manoeuvre time constant, Jeffreys support (h)	[0.1, 10]	[0.1, 10]
▶	Mach after an acceleration, uniform on (Mach)	[0.7, 0.87]	[0.73, 0.84]
	Altitude after a vertical manoeuvre, uniform on (ft)	[25000, 43000]	[25000, 43000]
	Altitude step (ft)	1000	1000
	Bank angle in turns (deg)	15	15
	Mach rate during an acceleration (Mach/s)	0.00166667	0.00166667
	Vertical rate during a climb or descent (ft/s)	66.6667	66.6667
	Mean time to lateral-navigation reversion (s)	31162.2	31162.2
	Cruise integration step (s)	10	10
▶	Manoeuvre integration step (s)	5	1
	Prior		
	Prior epoch (UTC)	2014-03-07T18:01:49Z	2014-03-07T18:01:49Z
▶	Position standard deviation (NM)	0.5	0.4
	Track standard deviation (deg)	1	1
	BFO bias prior mean (Hz)	150	150
	BFO bias prior standard deviation (Hz)	25	25
▶	Initial Mach set point	-	[0.82, 0.03]
▶	Prior mean position and track	-	not tabulated
	Autopilot-mode prior weights	[[true heading, 1], [magnetic heading,…	equal over the five modes
	Measurement		
	BTO fixed offset (us)	495679	495679
	BTO noise sd, R1200 (us)	29 at 7 epochs	29
	BTO noise sd, anomalous R1200 (us)	43 at 2 epochs	43
▶	BTO noise sd, R600 (us)	63 at 1 epoch	62
	BFO noise sd (Hz)	7 at 11 epochs	7
	BFO bias treatment	constant per trajectory, Kalman-margina…	constant per trajectory, Kalman-marginalised
	Prior		
	BFO bias drift rate (Hz^2/s)	0	0
	Measurement		
	BTO bias treatment	-	fixed per-flight T_channel only; no bias st…
▶	Vertical rate in the BFO model	True	not modelled
	00:19 BFO	-	not used; believed unreliable
	Model choice		
▶	Weather source	config/../data/era5-wind-temperature.bin	ACCESS-G, 150-500 hPa (about 18,000-44,000…
▶	Magnetic declination source	config/../data/igrf14-declination.bin	NOAA
▶	Resampling scheme	one filter per autopilot mode combined…	randomised branching: weight >= eta branche…
	Manoeuvre-type dependence	-	turns, accelerations and vertical manoeuvre…
	Turn extent (deg)	-	uniform on -180 to +180
▶	Cost-index speed mode	-	implemented but unused in most experiments;…
▶	Fuel and endurance	{'factor_mean': 1.0004, 'factor_sd': 0.…	infinite fuel; constraints applied to the p…
	Sampler		
▶	Fuel proposal	endurance (deadline unix 1394237459, pr…	n/a (no fuel model)
	Model choice		
	Observation cutoff	m0019b	00:19 (BTO only at 00:19)

▶ marks a difference. Rows whose value is prose or a file identifier cannot be compared by string match; for those the reference file asserts the relationship explicitly and parameters.csv records why. Quantities held per epoch rather than per run — the BTO and BFO measurement standard deviations — are in data/satcom-observations.csv and are compared in the note column of the CSV.

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Where was MH370 at 00:19 UTC?

A reproduction of the DSTG Bayesian aircraft-position pdf (Davey et al. 2016, Fig. 10.3)

Core run I, family 'free flight'. In the early flight, the speed and the direction can change freely. The speed is Mach 0.45 to 0.87 until 18:40, then Mach 0.70 to 0.87, always between 210 and 330 knots indicated air speed. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight.

This report recreates the probability density function (pdf) of the aircraft's latitude at the final satellite handshake, 00:19 UTC on 8 March 2014, from the model published by Davey, Gordon, Holland, Rutten and Williams, Bayesian Methods in the Search for MH370 (Springer, 2016). The aircraft state is filtered from the penultimate radar point at 18:01:49 UTC through all Inmarsat burst timing (BTO) and burst frequency (BFO) measurements, using the published cruise, manoeuvre and measurement models.

With BTO and BFO the recreated pdf has median 36.83°S and mode 37.05°S, with 95% of probability between 30.44°S and 38.14°S. The published curve has median 37.53°S, mode 37.90°S and 95% range 34.90–39.38°S. The two densities overlap by 64%. This curve is centred 0.7° further north and carries 37% of probability in the published northern shoulder between 34.5°S and 36.5°S (25% in the book).

Relative to the same family in run D, the median moves 0.11° north, the 34.5–36.5°S shoulder holds 37% of probability (base 29%), and the two curves overlap by 78%.

The curve pools 8 independent replicates of 7,000,000 particles. Single replicates overlap each other by 46%–86% and their medians span 0.93°. Over all balanced partitions of the 8 replicates the split-half overlap averages only 83.0% (range 75.8% to 90.3%) against the 8-replicate floor of 0.924, so the pooled curve is NOT converged at this particle count and quantities read off it carry that caveat.

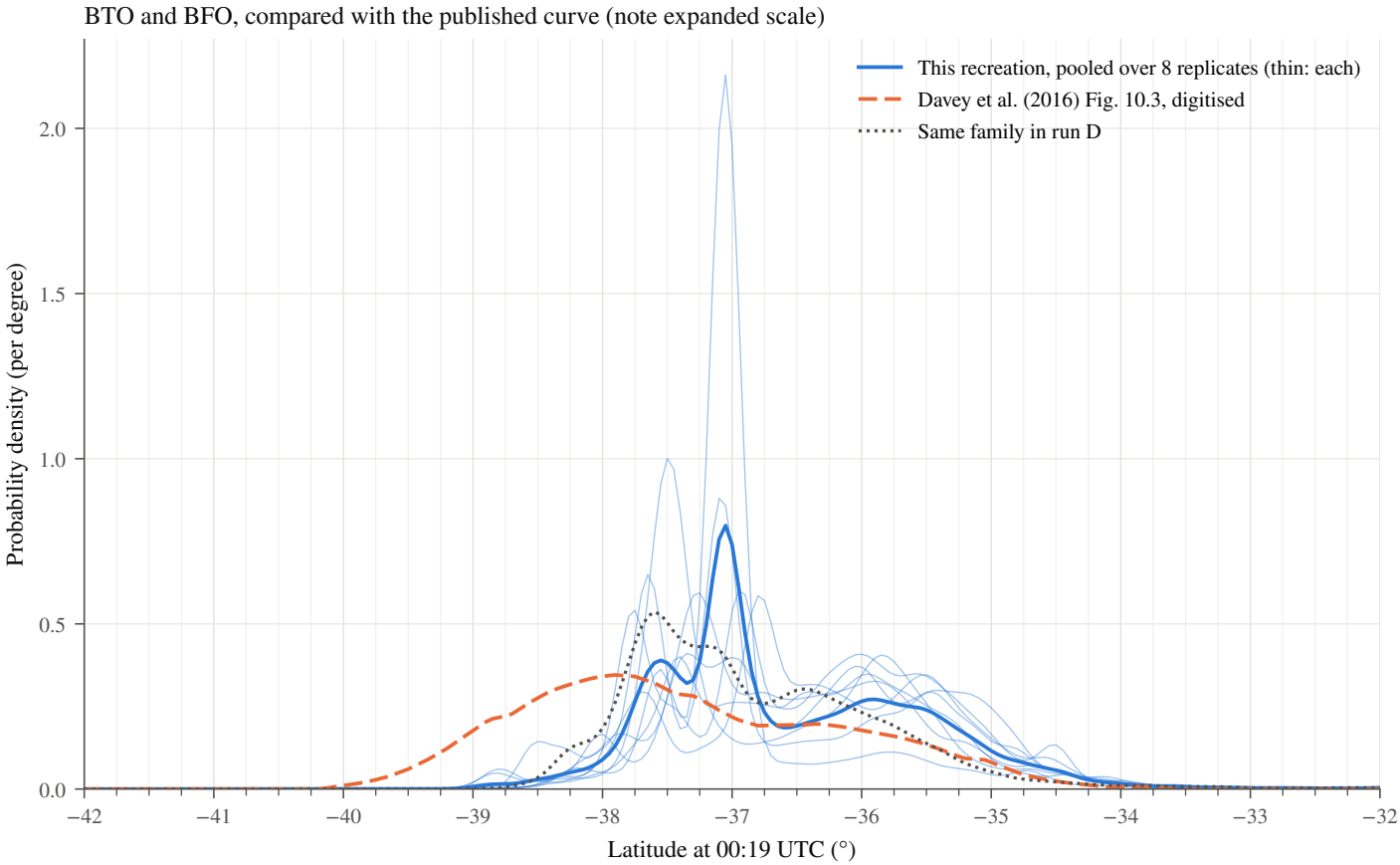


Figure 1. Latitude pdf of the aircraft at 00:19:37 UTC, the form of Davey et al. Fig. 10.3. BTO and BFO, with the published curve. Densities are weighted particle histograms smoothed with a 0.1° Gaussian; each particle carries its own altitude. Evidence: BTO at every epoch except the 18:39 and 23:15 C-channel calls; BFO (bottom only) at 18:28, 18:39, 19:41–22:41, 23:15 and 00:11. This run also uses: primary-radar positions and track segments 18:04-18:22 scored as measurements; fuel model internal-v1, two tanks, one-engine flight after the first flame-out (hold-taper drift-down), paths dry before 00:11 rejected. No debris, drift, search or end-of-flight evidence is used. Run conditions as stated at the top of this page.

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Technical: run next-i-free; stack: run D stack + speed rule (see the method page); binary 1ff4d1d, x86_64-linux (deskstar); seeds 1-8 x 7,000,000 particles; ln Z -135.78; split-half mean over 35 balanced partitions 0.830 (range 0.758-0.903) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.

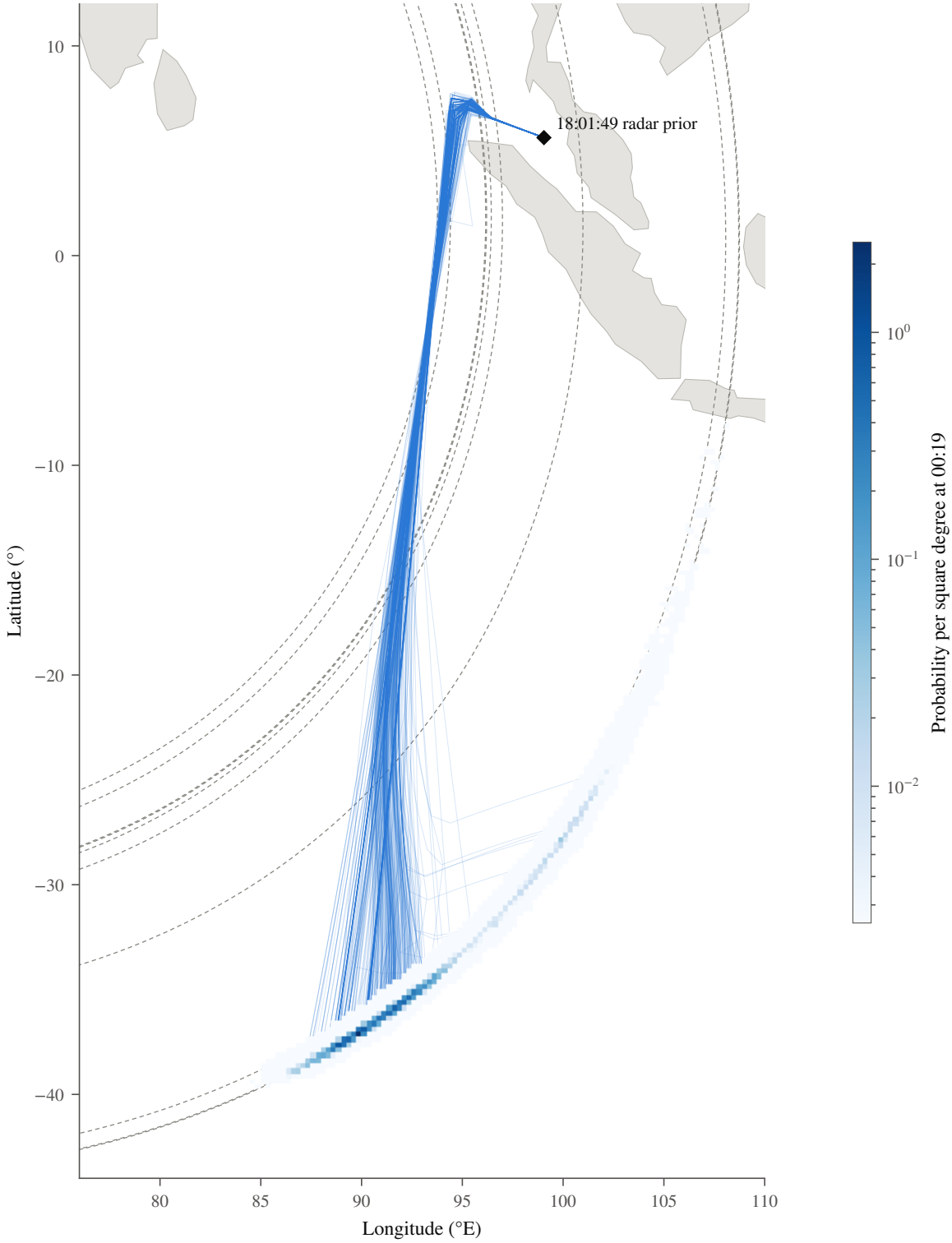
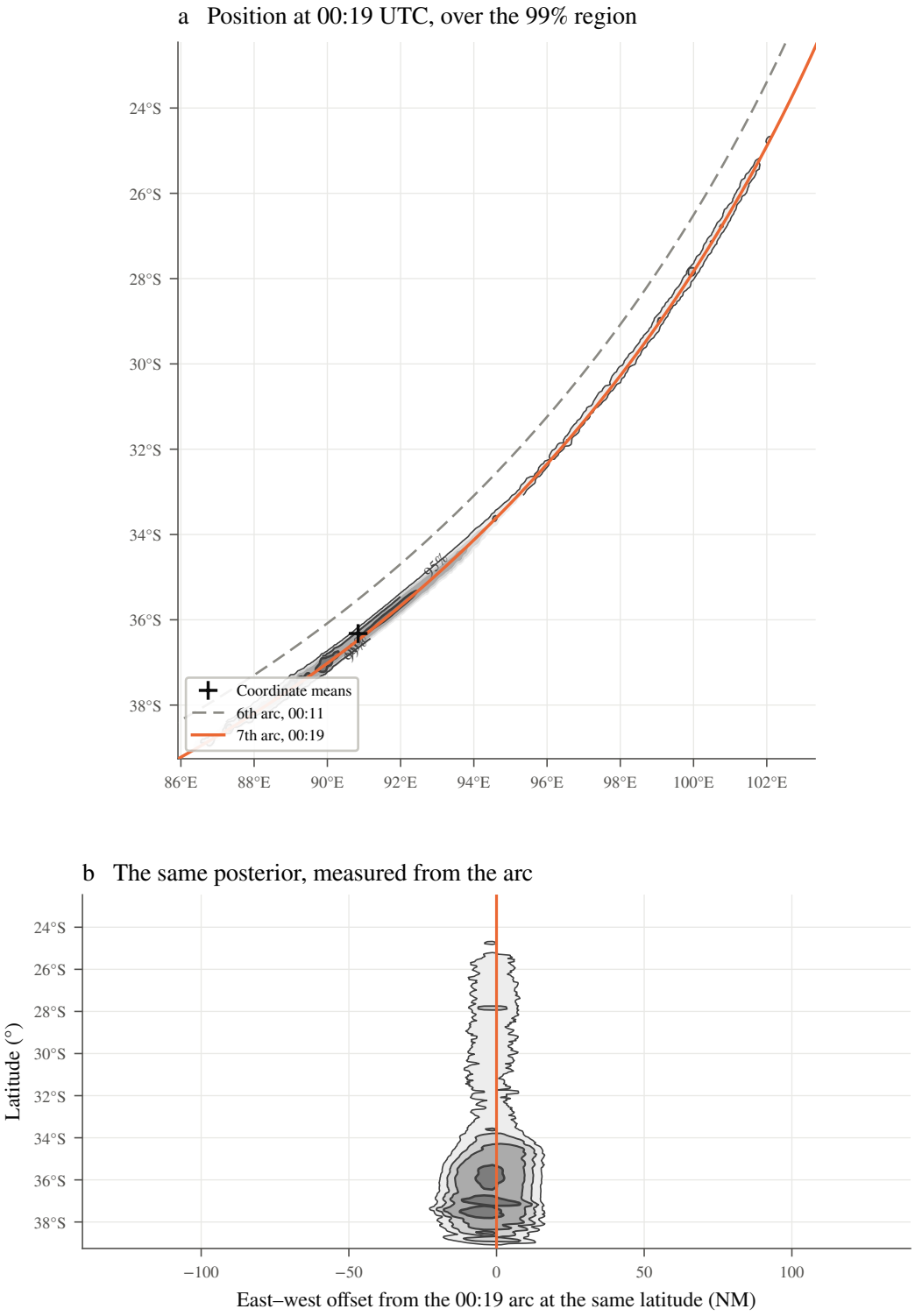


Figure 2. Posterior aircraft position at 00:19 UTC with BTO and BFO, pooled over 8 replicates (blue shading, logarithmic), and 400 routes drawn in proportion to posterior weight from replicate seed 1 (thin lines, 10-minute vertices). Dashed grey curves are the BTO arcs at a reference altitude of 35,000 ft; they are drawn for orientation only. Land: Natural Earth 1:110m.

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DIFFERENCES FROM DAVEY

Mach after an acceleration, uniform on: [0.7, 0.87]

Manoeuvre integration step: 5

Position sd: 0.5

Initial Mach set point: -

Prior mean position and track:

BTO noise sd, R600: 63 at 1 epoch

Vertical rate in the BFO model: True

Weather source: era5-wind-temperature.bin

Magnetic declination source: igrf14-declination.bin

Resampling scheme: one filter per autopilot …

Cost-index speed mode:

Fuel and endurance: {'factor_mean': 1.0004, '…

Fuel proposal: endurance (deadline unix …

book values: see the parameter page

POSTERIOR AT 00:19

median 36.83°S, mode 37.05°S

95% 30.44–38.14°S

replicates 8, split-half 0.734

largest single draw 0.00% of weight

99% region 18,627 NM²

95% region 8,970 NM²

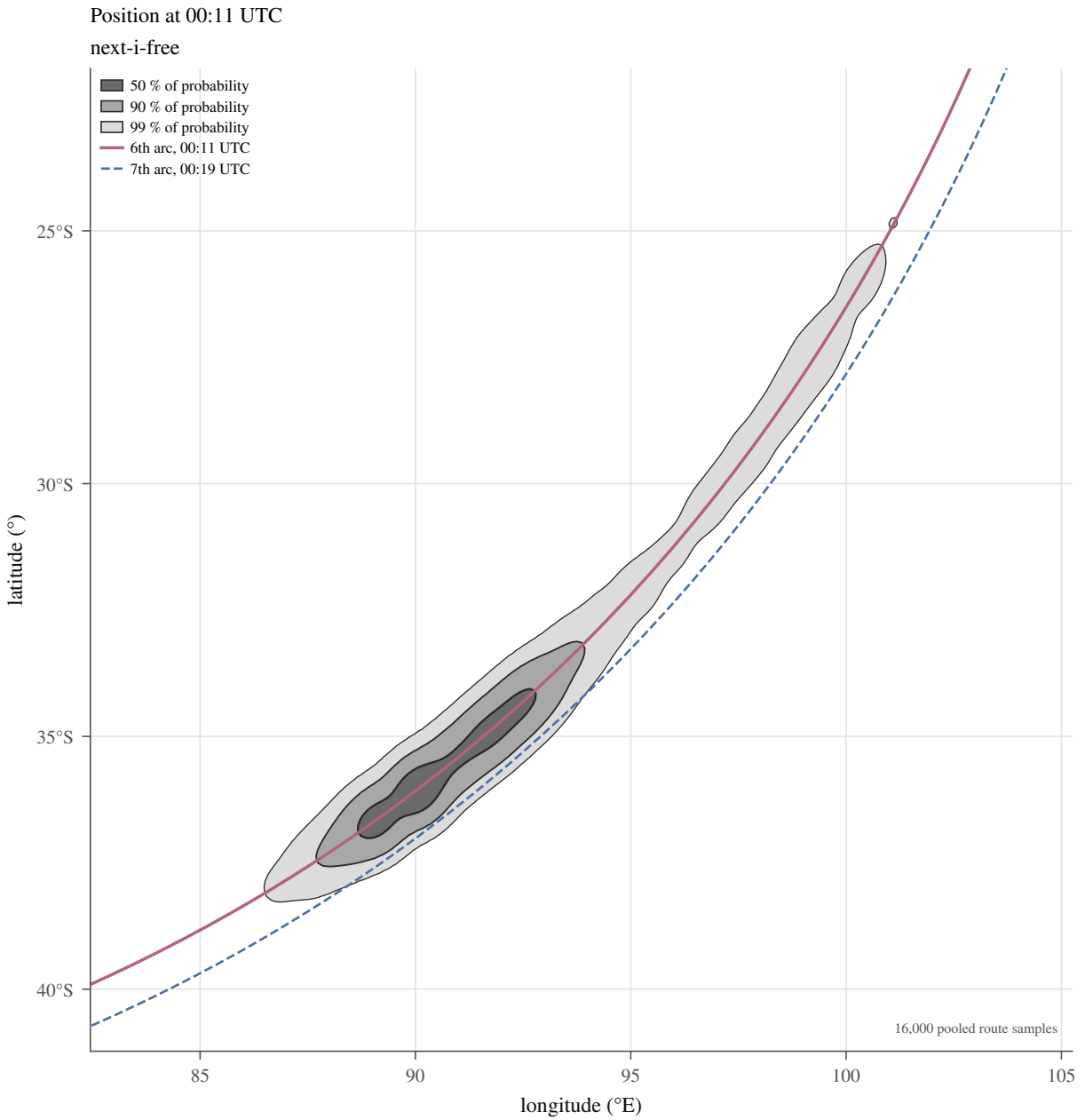
90% region 5,964 NM²

50% region 1,422 NM²

Figure 3. Joint latitude-longitude posterior at 00:19 UTC, drawn as highest-posterior-density regions: the 50% curve bounds the smallest area holding 50% of the probability, and so on outward, so a region here is the smallest place to look for that confidence rather than an interval in either coordinate alone. Binned at 0.05° and smoothed over three bins. Panel a spans the bounding box of the 99% region, so sparse outlying cells are excluded; enclosed areas (NM², at the view's mean latitude) are 99% 18,627, 95% 8,970, 90% 5,964, 50% 1,422. Panel b replots the same particles against their east-west offset from the 00:19 arc, because the region is about 0.3° wide and 17° long and the inner contours are not separable at a true aspect ratio. Arcs are the reference BTO loci at 35,000 ft; 00:19 is the 7th. Land: Natural Earth 1:110m.

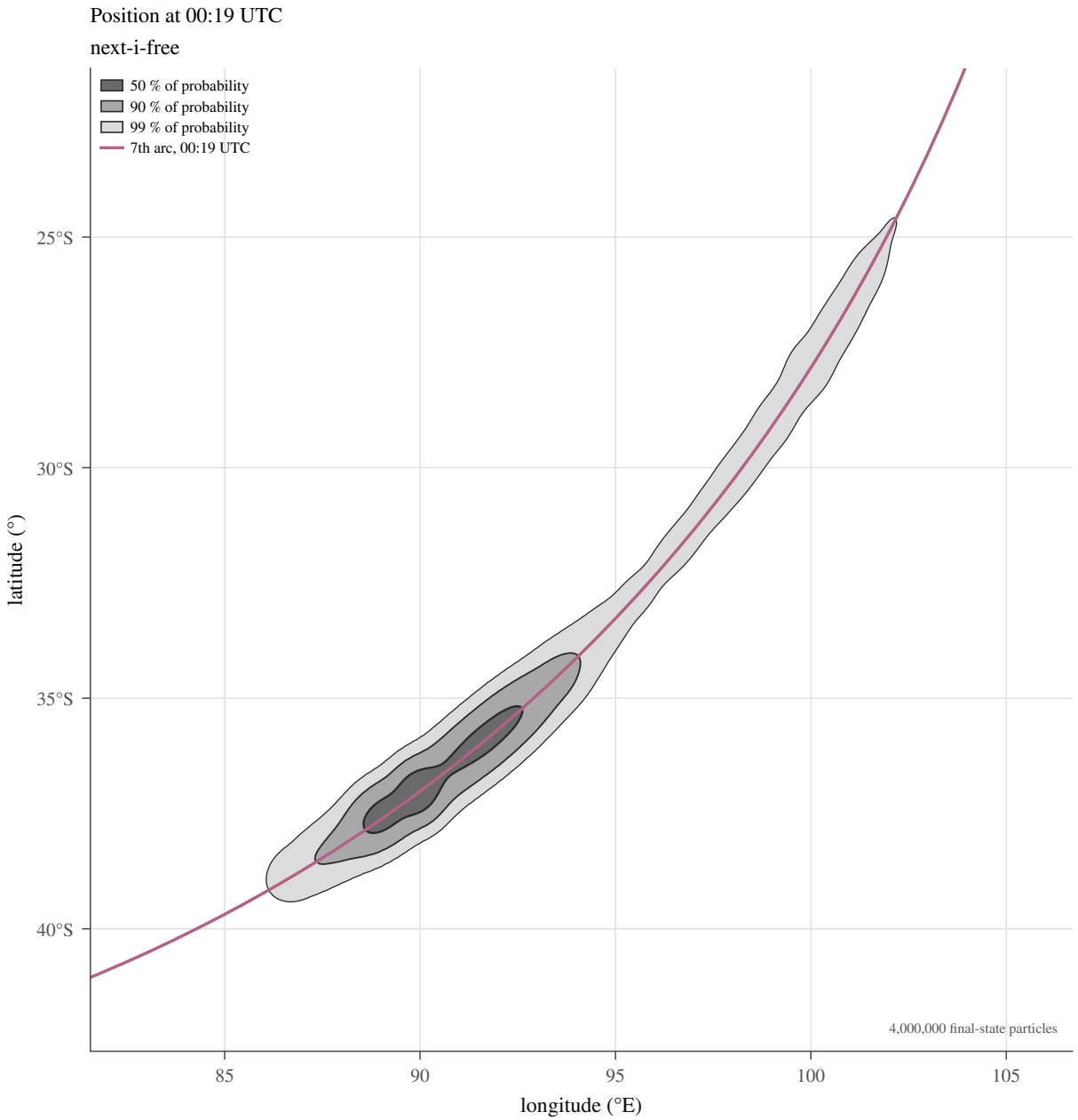
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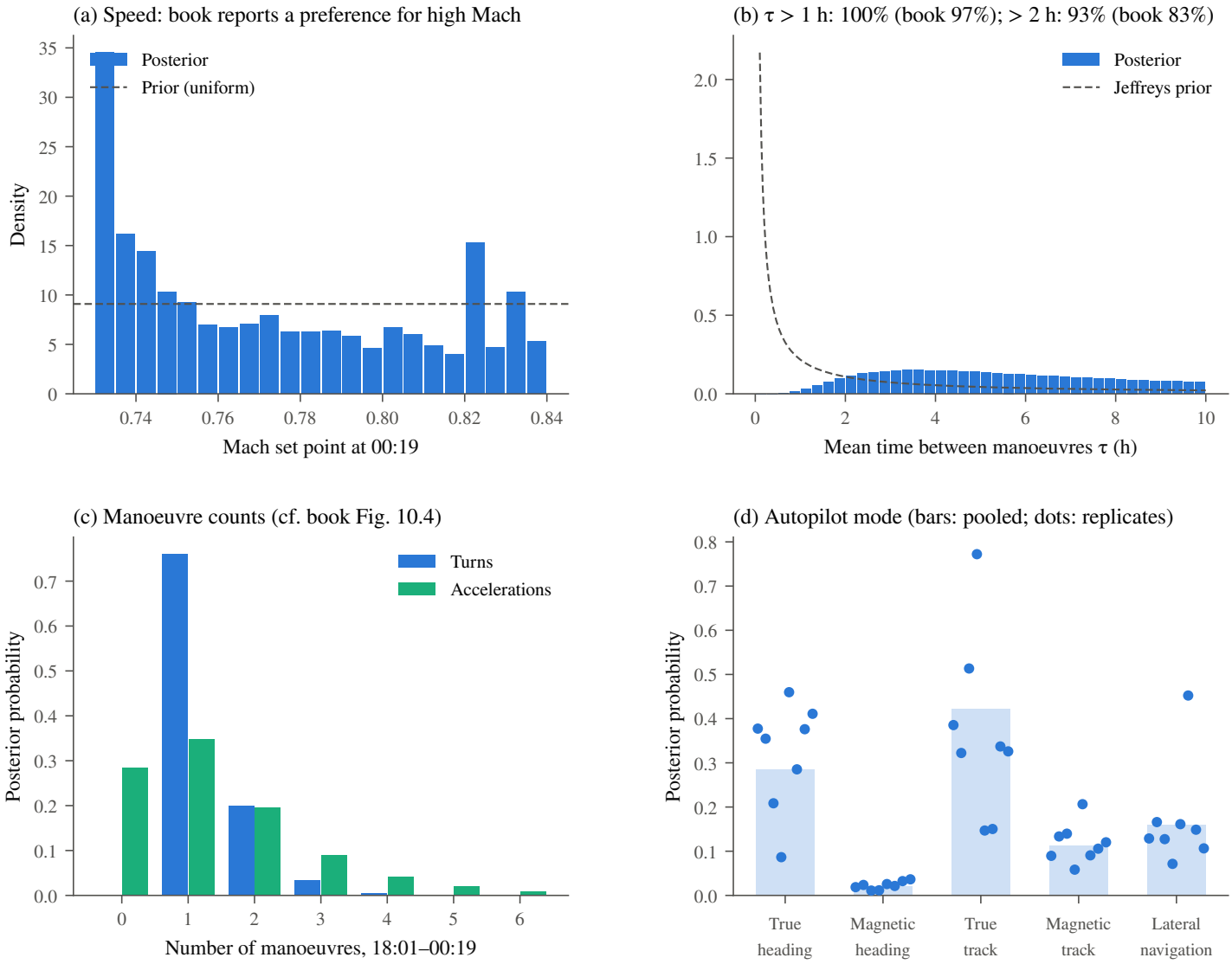
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Posterior behaviour compared with the book



	This recreation	Davey et al. (2016)
Median latitude	36.83°S	37.53°S
Mode	37.05°S	37.90°S
95% interval	30.44–38.14°S	34.90–39.38°S
95% width	7.70°	4.47°
Overlap with published pdf	64% (replicates 46%–66%)	—
Median Mach at 00:19	0.766	upper part of 0.73–0.84

Figure 3 and Table 1. Posterior summaries with BTO and BFO, pooled over replicates, alongside the statements in Davey et al. (2016), sec. 10.2–10.4. Book values of the latitude pdf come from the digitised Fig. 10.3 (bottom); other book values are quoted from the text.

Plain: Core run I, family 'free flight'. In the early flight, the speed and the direction can change freely. The speed is Mach 0.45 to 0.87 until 18:40, then Mach 0.70 to 0.87, always between 210 and 330 knots indicated air speed. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.830, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.
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Method, convergence and reproduction

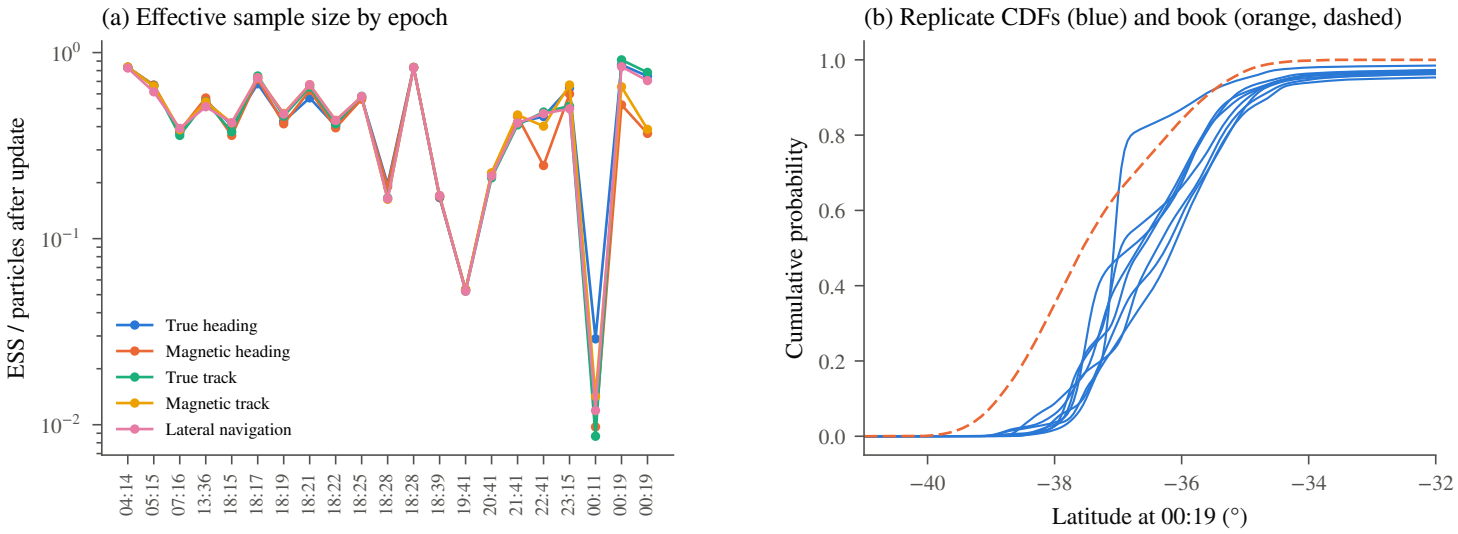


Figure 4. (a) Effective sample size after each measurement update, averaged over replicates, per autopilot-mode filter. The narrowest bottlenecks are the 18:39 C-channel BFO (the southward turn) and the 19:41 BTO arc. (b) Latitude CDFs of the 8 independent replicates; the largest difference between any two is 0.493 (Kolmogorov distance).

Model. The state, dynamics and likelihood follow Davey et al. (2016) chapters 4–8: prior at 18:01:49 UTC with 0.5 NM position and 1° track uncertainty, Mach uniform on 0.73–0.84, altitude on 1,000 ft levels from 25,000 to 43,000 ft; five autopilot modes (true/magnetic heading, true/magnetic track, lateral navigation with a one-off reversion to heading hold); Ornstein–Uhlenbeck fluctuations of Mach, control angle and wind error with the published rates; turns, speed changes and climbs arriving with exponential gaps of common mean τ (Jeffreys prior on 0.1–10 h) at 15° bank, 0.1 Mach per minute and 4,000 ft per minute. BTO is Gaussian with the tabulated per-message errors (29, 43 or 63 µs); BFO has 7 Hz noise and an unknown constant bias (prior 150 ± 25 Hz) marginalised exactly per particle with a Kalman filter.

Substitutions. The book's ACCESS-G weather is replaced by ERA5 winds and temperatures; magnetic declination is IGRF-14. The prior mean position and track are reconstructed from the book's radar figures because they are not tabulated. Davey et al. used variable-rate branching; this recreation uses fixed-size particle filters with systematic resampling when ESS falls below 50% of the particles. The book warns that conventional resampling collapses static parameters (sec. 10.4). Both are therefore handled exactly: one filter per autopilot mode, combined in proportion to its estimated marginal likelihood, and a Gibbs refresh of τ from its path conditional after every resampling step. Without them, a plain bootstrap filter of one million particles retained only about one hundred distinct prior draws, and an earlier recreation in this project found tails and turn counts that changed with the random seed.

Run. Configuration next-i-free; particles per mode 1,000,000, 500,000, 2,500,000, 500,000, 2,500,000 (order as in Figure 3d), 7,000,000 per replicate; BTO+BFO seeds 1, 2, 3, 4, 5, 6, 7, 8; code revision 1ff4d1d; 88 threads; total runtime 100.1 min; peak memory 30,347 MiB. Hypotheses enabled: radar-fix (parameters in run.json). Sensitivity run 'next-i-free', varying cas_envelope_kt (not declared) to [210.0, 330.0]; mach_range [0.73, 0.84] to [0.7, 0.87]; replicates 4 to 8.

Reproduce with one command from the repository root: make hypothesis H=radar-fix

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Parameters and model choices against Davey et al. (2016)

43 parameters and choices. 30 match the published model; 13 differ and are marked. A difference is not by itself a defect: some are deliberate and documented (ERA5 for the unavailable ACCESS-G, IGRF-14 for NOAA, the sampler), some are the subject of the run (a widened Mach range), and some are worth fixing. Page numbers refer to the book; the full table with citations and notes is written beside this report as parameters.csv.

	Parameter	This run	Davey et al. (2016)
	Dynamics		
	Mach OU reversion rate (1/s)	0.01058	0.01058
	Mach OU noise strength (1/s)	2.05e-07	2.05e-07
	Control-angle OU reversion rate (1/s)	0.009792	0.009792
	Control-angle OU noise strength (rad^2/s)	4.074e-08	4.074e-08
	Wind-error OU reversion rate (1/s)	0.001087	0.001087
	Wind-error OU noise strength (kn^2/s)	0.07021	0.07021
	Manoeuvre time constant, Jeffreys support (h)	[0.1, 10]	[0.1, 10]
▶	Mach after an acceleration, uniform on (Mach)	[0.7, 0.87]	[0.73, 0.84]
	Altitude after a vertical manoeuvre, uniform on (ft)	[25000, 43000]	[25000, 43000]
	Altitude step (ft)	1000	1000
	Bank angle in turns (deg)	15	15
	Mach rate during an acceleration (Mach/s)	0.00166667	0.00166667
	Vertical rate during a climb or descent (ft/s)	66.6667	66.6667
	Mean time to lateral-navigation reversion (s)	31162.2	31162.2
	Cruise integration step (s)	10	10
▶	Manoeuvre integration step (s)	5	1
	Prior		
	Prior epoch (UTC)	2014-03-07T18:01:49Z	2014-03-07T18:01:49Z
▶	Position standard deviation (NM)	0.5	0.4
	Track standard deviation (deg)	1	1
	BFO bias prior mean (Hz)	150	150
	BFO bias prior standard deviation (Hz)	25	25
▶	Initial Mach set point	-	[0.82, 0.03]
▶	Prior mean position and track	-	not tabulated
	Autopilot-mode prior weights	[[true heading, 1], [magnetic heading,…	equal over the five modes
	Measurement		
	BTO fixed offset (us)	495679	495679
	BTO noise sd, R1200 (us)	29 at 7 epochs	29
	BTO noise sd, anomalous R1200 (us)	43 at 2 epochs	43
▶	BTO noise sd, R600 (us)	63 at 1 epoch	62
	BFO noise sd (Hz)	7 at 11 epochs	7
	BFO bias treatment	constant per trajectory, Kalman-margina…	constant per trajectory, Kalman-marginalised
	Prior		
	BFO bias drift rate (Hz^2/s)	0	0
	Measurement		
	BTO bias treatment	-	fixed per-flight T_channel only; no bias st…
▶	Vertical rate in the BFO model	True	not modelled
	00:19 BFO	-	not used; believed unreliable
	Model choice		
▶	Weather source	config/../data/era5-wind-temperature.bin	ACCESS-G, 150-500 hPa (about 18,000-44,000…
▶	Magnetic declination source	config/../data/igrf14-declination.bin	NOAA
▶	Resampling scheme	one filter per autopilot mode combined…	randomised branching: weight >= eta branche…
	Manoeuvre-type dependence	-	turns, accelerations and vertical manoeuvre…
	Turn extent (deg)	-	uniform on -180 to +180
▶	Cost-index speed mode	-	implemented but unused in most experiments;…
▶	Fuel and endurance	{'factor_mean': 1.0004, 'factor_sd': 0.…	infinite fuel; constraints applied to the p…
	Sampler		
▶	Fuel proposal	endurance (deadline unix 1394237459, pr…	n/a (no fuel model)
	Model choice		
	Observation cutoff	m0019b	00:19 (BTO only at 00:19)

▶ marks a difference. Rows whose value is prose or a file identifier cannot be compared by string match; for those the reference file asserts the relationship explicitly and parameters.csv records why. Quantities held per epoch rather than per run — the BTO and BFO measurement standard deviations — are in data/satcom-observations.csv and are compared in the note column of the CSV.

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Technical: run next-i-free; stack: run D stack + speed rule (see the method page); binary 1ff4d1d, x86_64-linux (deskstar); seeds 1-8 x 7,000,000 particles; ln Z -135.78; split-half mean over 35 balanced partitions 0.830 (range 0.758-0.903) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.

Where was MH370 at 00:19 UTC?

A reproduction of the DSTG Bayesian aircraft-position pdf (Davey et al. 2016, Fig. 10.3)

Core run I, family 'Davey dynamics'. In the early flight, the aircraft moves as in the Davey model, with Mach 0.73 to 0.84 throughout. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight.

This report recreates the probability density function (pdf) of the aircraft's latitude at the final satellite handshake, 00:19 UTC on 8 March 2014, from the model published by Davey, Gordon, Holland, Rutten and Williams, Bayesian Methods in the Search for MH370 (Springer, 2016). The aircraft state is filtered from the penultimate radar point at 18:01:49 UTC through all Inmarsat burst timing (BTO) and burst frequency (BFO) measurements, using the published cruise, manoeuvre and measurement models.

With BTO and BFO the recreated pdf has median 36.73°S and mode 37.45°S, with 95% of probability between 27.40°S and 38.31°S. The published curve has median 37.53°S, mode 37.90°S and 95% range 34.90–39.38°S. The two densities overlap by 73%. This curve is centred 0.8° further north and carries 35% of probability in the published northern shoulder between 34.5°S and 36.5°S (25% in the book).

Relative to the same family in run D, the median moves 0.25° south, the 34.5–36.5°S shoulder holds 35% of probability (base 40%), and the two curves overlap by 90%.

The curve pools 8 independent replicates of 3,500,000 particles. Single replicates overlap each other by 65%–89% and their medians span 1.21°. Over all balanced partitions of the 8 replicates the split-half overlap averages only 89.1% (range 85.3% to 92.7%) against the 8-replicate floor of 0.924, so the pooled curve is NOT converged at this particle count and quantities read off it carry that caveat.

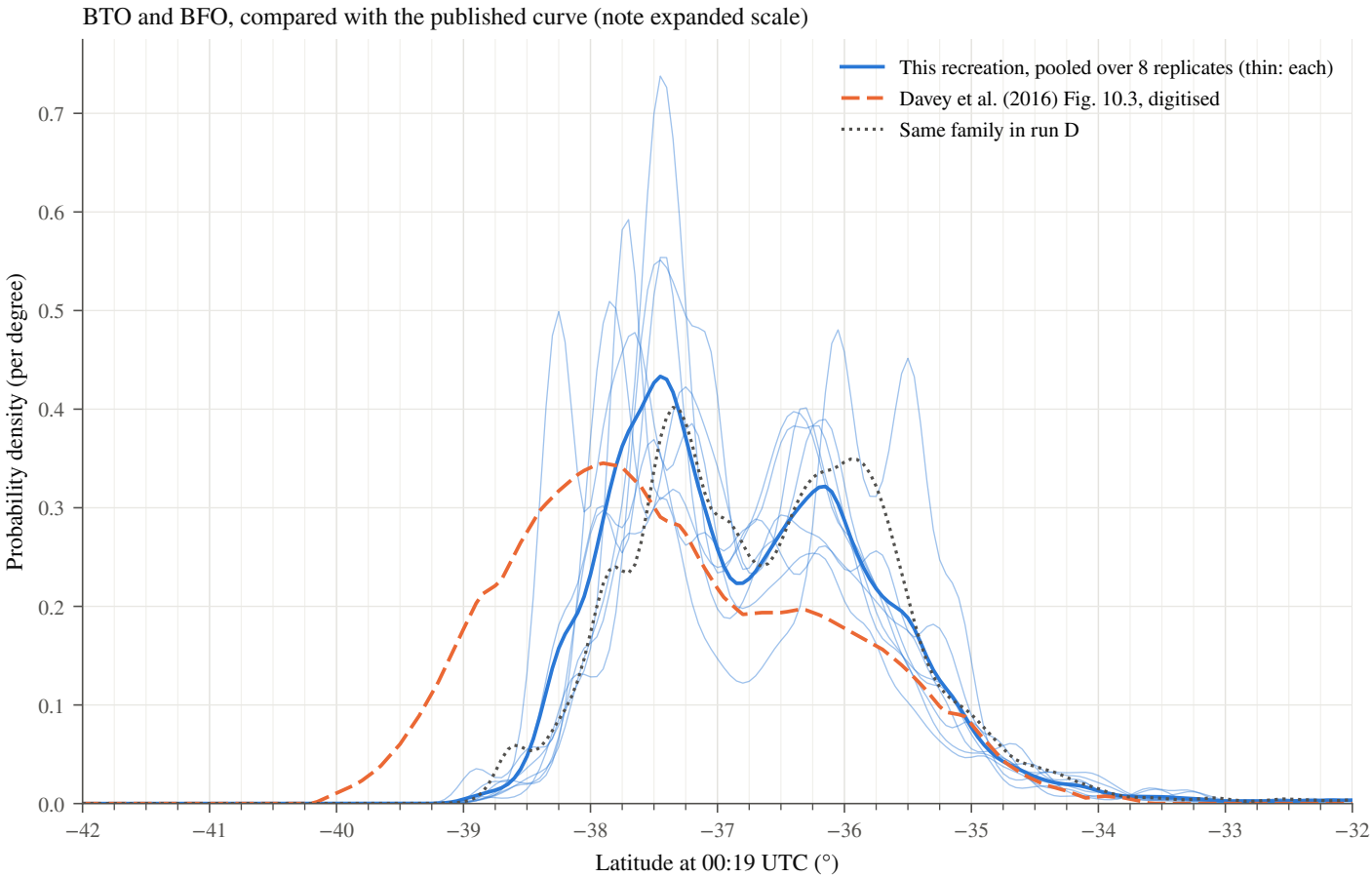


Figure 1. Latitude pdf of the aircraft at 00:19:37 UTC, the form of Davey et al. Fig. 10.3. BTO and BFO, with the published curve. Densities are weighted particle histograms smoothed with a 0.1° Gaussian; each particle carries its own altitude. Evidence: BTO at every epoch except the 18:39 and 23:15 C-channel calls; BFO (bottom only) at 18:28, 18:39, 19:41–22:41, 23:15 and 00:11. This run also uses: primary-radar positions and track segments 18:04-18:22 scored as measurements; fuel model internal-v1, two tanks, one-engine flight after the first flame-out (hold-taper drift-down), paths dry before 00:11 rejected. No debris, drift, search or end-of-flight evidence is used. Hypotheses enabled: radar-fix (parameters in run.json). Sensitivity run 'next-i-repro-radar', varying replicates 4 to 8.

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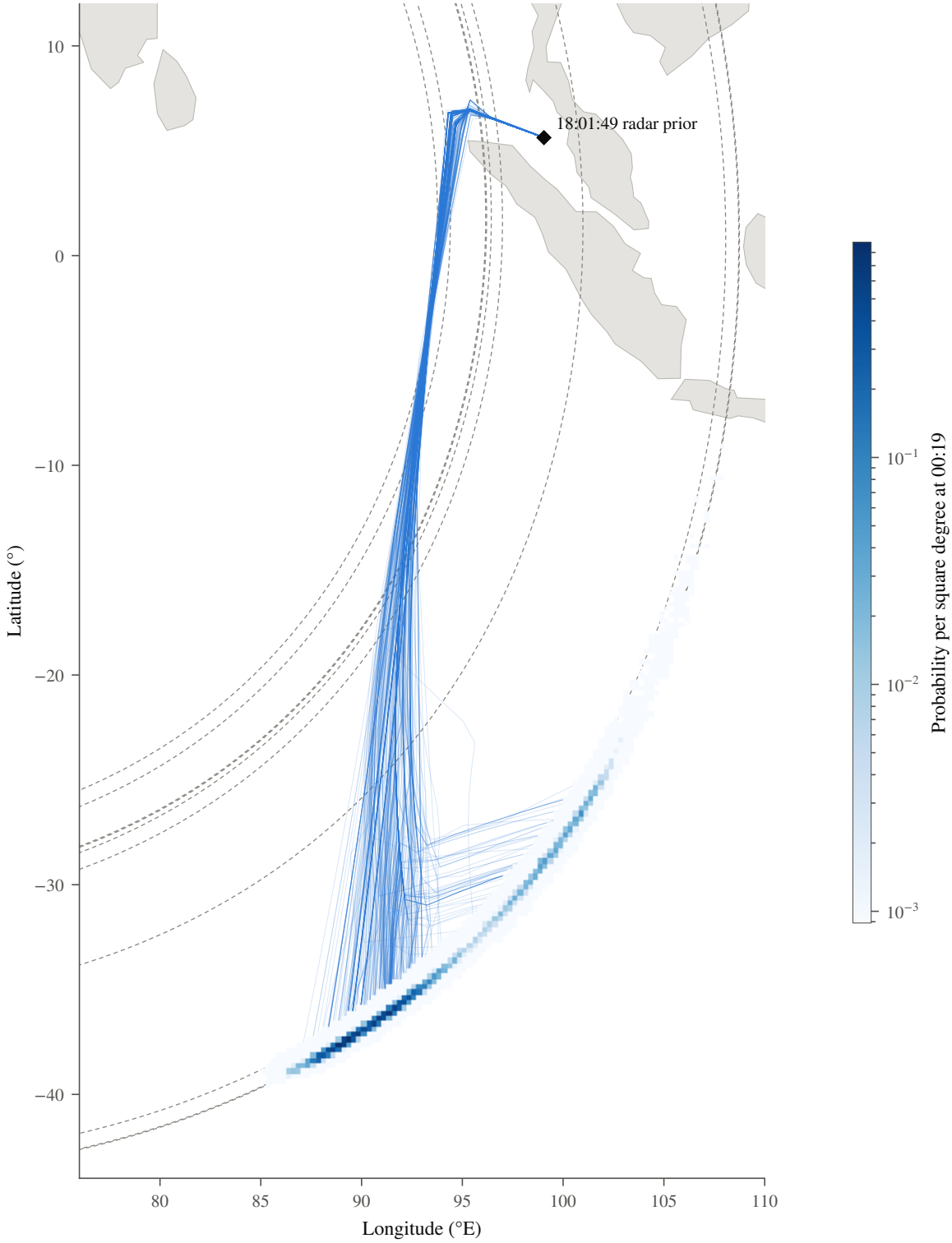
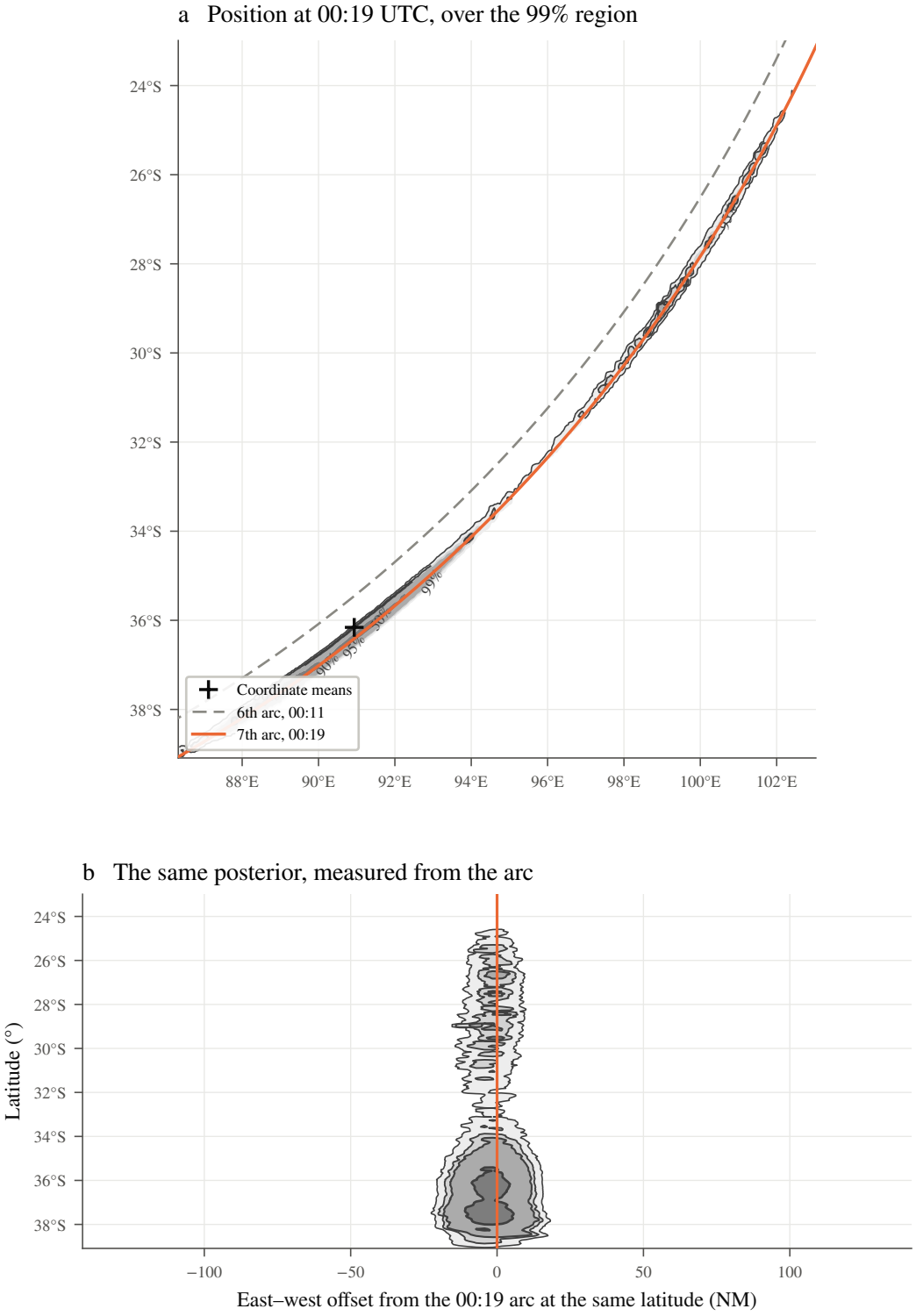


Figure 2. Posterior aircraft position at 00:19 UTC with BTO and BFO, pooled over 8 replicates (blue shading, logarithmic), and 400 routes drawn in proportion to posterior weight from replicate seed 1 (thin lines, 10-minute vertices). Dashed grey curves are the BTO arcs at a reference altitude of 35,000 ft; they are drawn for orientation only. Land: Natural Earth 1:110m.

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DIFFERENCES FROM DAVEY

Manoeuvre integration step: 5

Position sd: 0.5

Initial Mach set point: -

Prior mean position and track:

BTO noise sd, R600: 63 at 1 epoch

Vertical rate in the BFO model: True

Weather source: era5-wind-temperature.bin

Magnetic declination source: igrf14-declination.bin

Resampling scheme: one filter per autopilot …

Cost-index speed mode:

Fuel and endurance: {'factor_mean': 1.0004, '…

Fuel proposal: endurance (deadline unix …

book values: see the parameter page

POSTERIOR AT 00:19

median 36.73°S, mode 37.45°S

95% 27.40–38.31°S

replicates 8, split-half 0.864

largest single draw 0.00% of weight

99% region 20,381 NM²

95% region 11,891 NM²

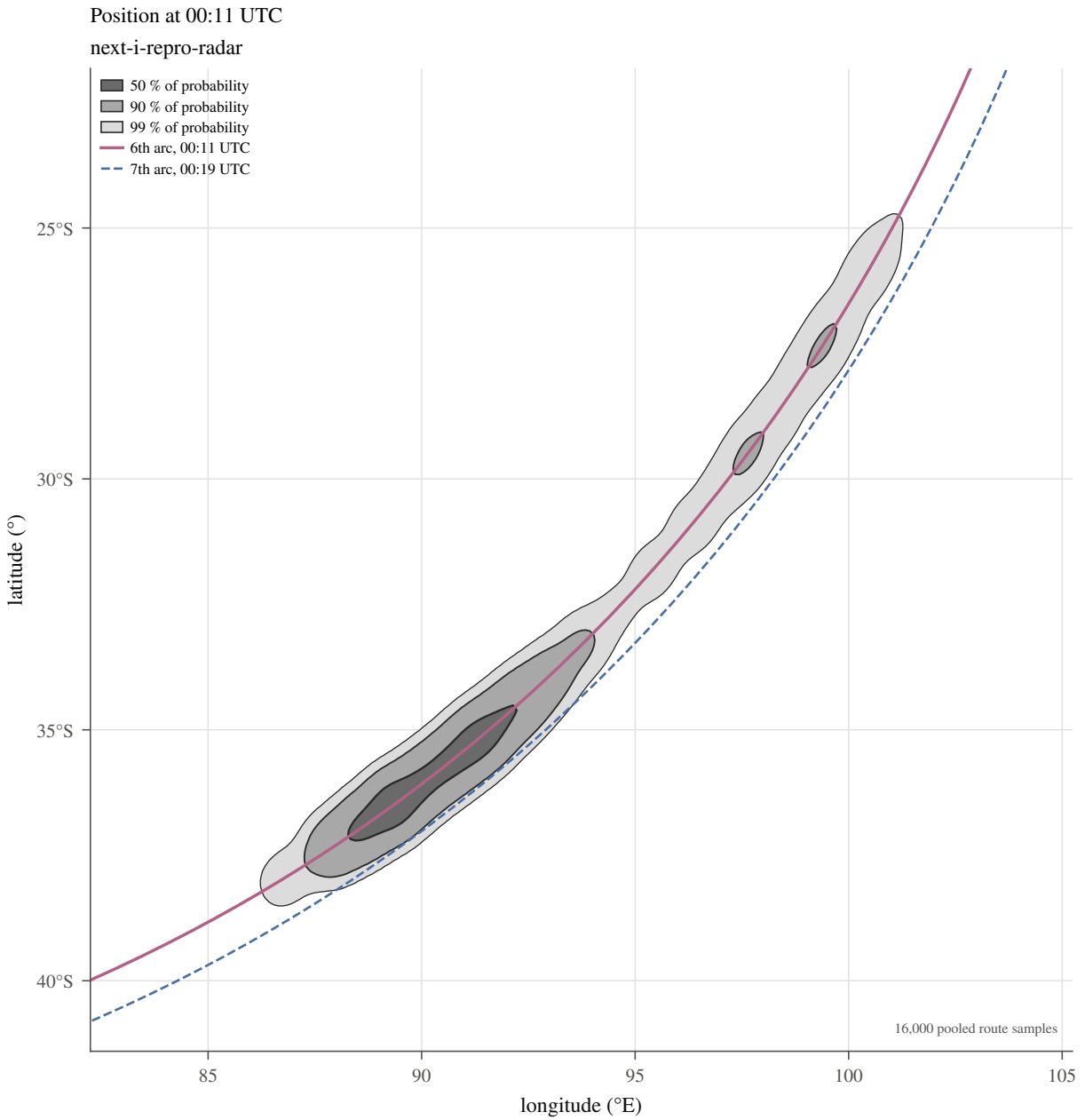
90% region 7,827 NM²

50% region 1,974 NM²

Figure 3. Joint latitude-longitude posterior at 00:19 UTC, drawn as highest-posterior-density regions: the 50% curve bounds the smallest area holding 50% of the probability, and so on outward, so a region here is the smallest place to look for that confidence rather than an interval in either coordinate alone. Binned at 0.05° and smoothed over three bins. Panel a spans the bounding box of the 99% region, so sparse outlying cells are excluded; enclosed areas (NM², at the view's mean latitude) are 99% 20,381, 95% 11,891, 90% 7,827, 50% 1,974. Panel b replots the same particles against their east-west offset from the 00:19 arc, because the region is about 0.3° wide and 16° long and the inner contours are not separable at a true aspect ratio. Arcs are the reference BTO loci at 35,000 ft; 00:19 is the 7th. Land: Natural Earth 1:110m.

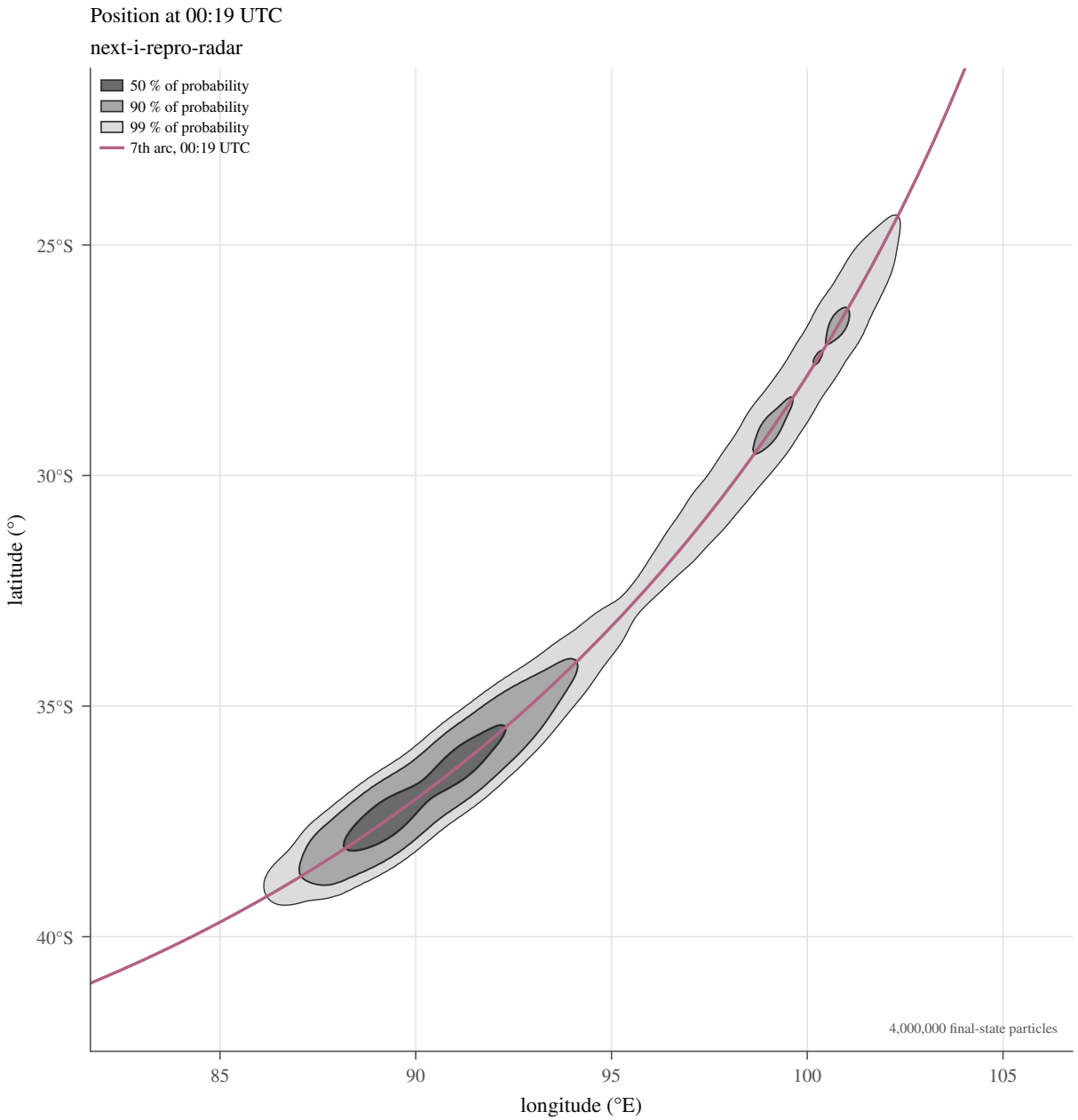
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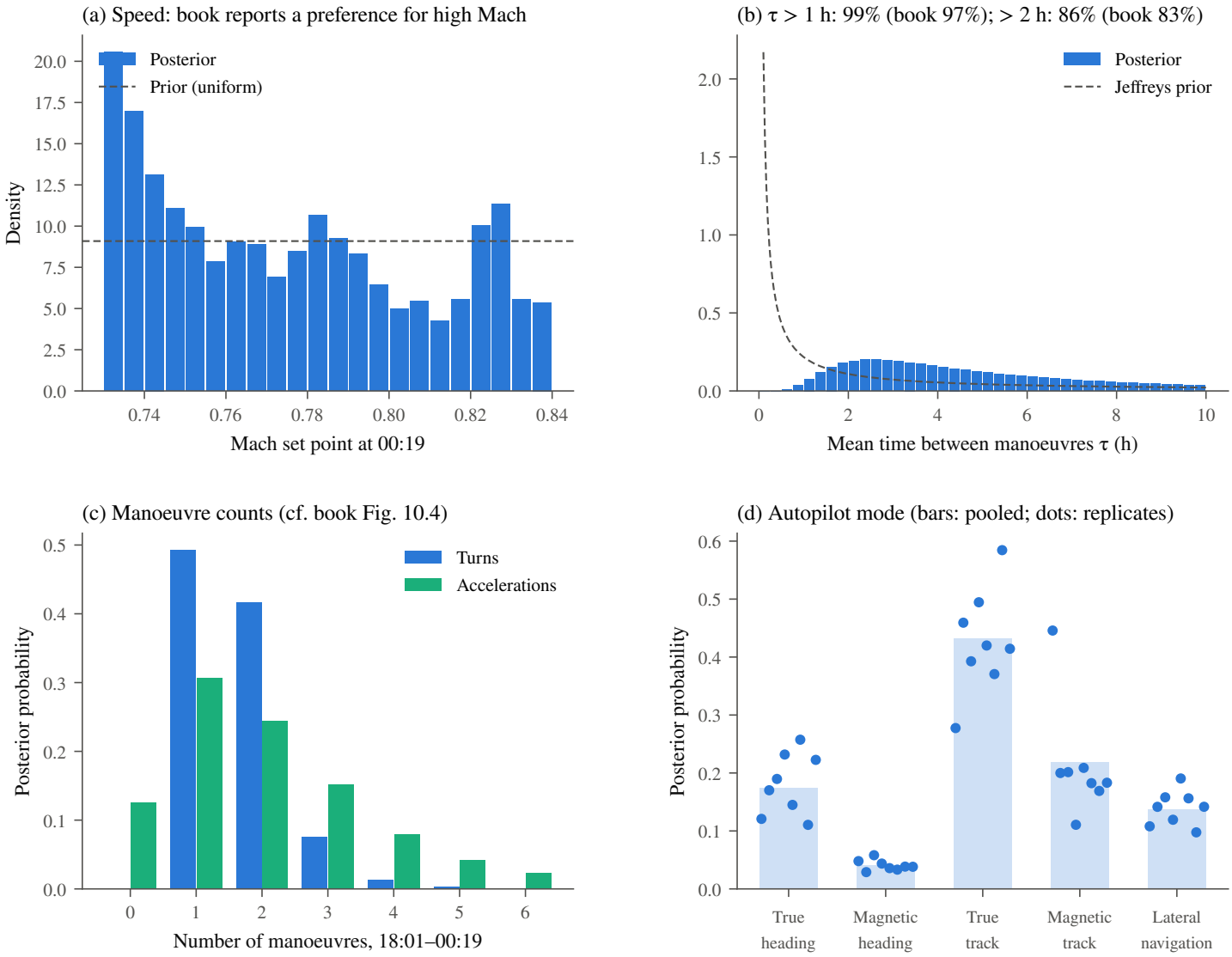
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	This recreation	Davey et al. (2016)
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Mode	37.45°S	37.90°S
95% interval	27.40–38.31°S	34.90–39.38°S
95% width	10.91°	4.47°
Overlap with published pdf	73% (replicates 60%–79%)	—
Median Mach at 00:19	0.772	upper part of 0.73–0.84

Figure 3 and Table 1. Posterior summaries with BTO and BFO, pooled over replicates, alongside the statements in Davey et al. (2016), sec. 10.2–10.4. Book values of the latitude pdf come from the digitised Fig. 10.3 (bottom); other book values are quoted from the text.

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Method, convergence and reproduction

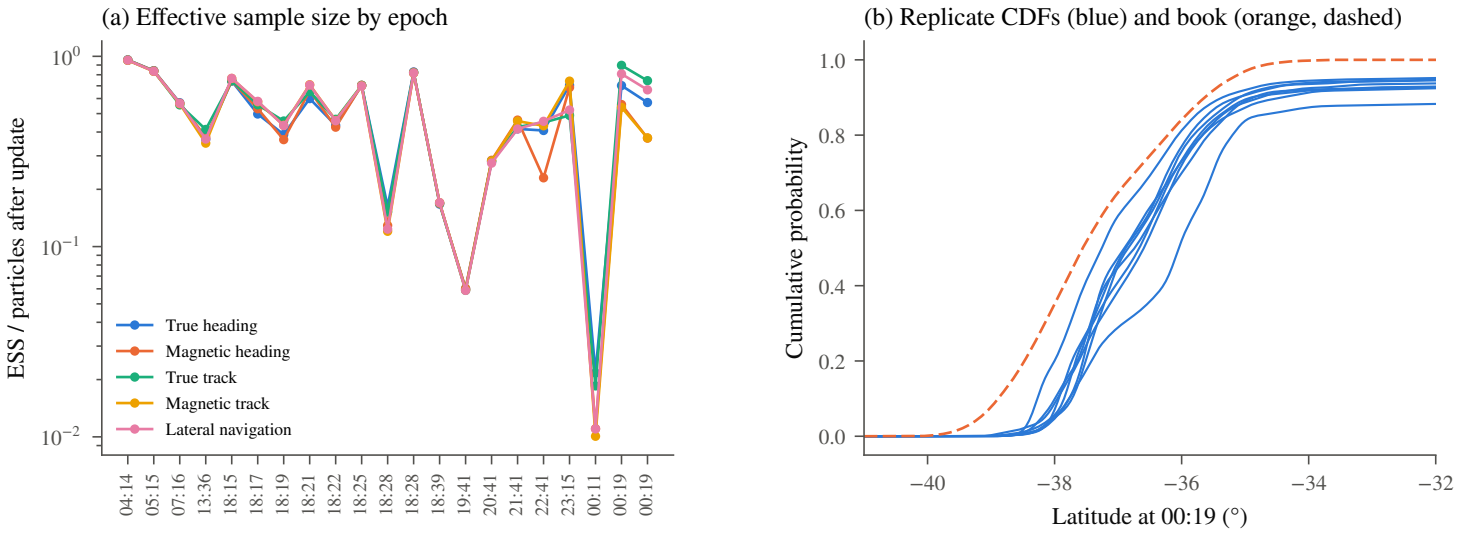


Figure 4. (a) Effective sample size after each measurement update, averaged over replicates, per autopilot-mode filter. The narrowest bottlenecks are the 18:39 C-channel BFO (the southward turn) and the 19:41 BTO arc. (b) Latitude CDFs of the 8 independent replicates; the largest difference between any two is 0.344 (Kolmogorov distance).

Model. The state, dynamics and likelihood follow Davey et al. (2016) chapters 4–8: prior at 18:01:49 UTC with 0.5 NM position and 1° track uncertainty, Mach uniform on 0.73–0.84, altitude on 1,000 ft levels from 25,000 to 43,000 ft; five autopilot modes (true/magnetic heading, true/magnetic track, lateral navigation with a one-off reversion to heading hold); Ornstein–Uhlenbeck fluctuations of Mach, control angle and wind error with the published rates; turns, speed changes and climbs arriving with exponential gaps of common mean τ (Jeffreys prior on 0.1–10 h) at 15° bank, 0.1 Mach per minute and 4,000 ft per minute. BTO is Gaussian with the tabulated per-message errors (29, 43 or 63 µs); BFO has 7 Hz noise and an unknown constant bias (prior 150 ± 25 Hz) marginalised exactly per particle with a Kalman filter.

Substitutions. The book's ACCESS-G weather is replaced by ERA5 winds and temperatures; magnetic declination is IGRF-14. The prior mean position and track are reconstructed from the book's radar figures because they are not tabulated. Davey et al. used variable-rate branching; this recreation uses fixed-size particle filters with systematic resampling when ESS falls below 50% of the particles. The book warns that conventional resampling collapses static parameters (sec. 10.4). Both are therefore handled exactly: one filter per autopilot mode, combined in proportion to its estimated marginal likelihood, and a Gibbs refresh of τ from its path conditional after every resampling step. Without them, a plain bootstrap filter of one million particles retained only about one hundred distinct prior draws, and an earlier recreation in this project found tails and turn counts that changed with the random seed.

Run. Configuration next-i-repro-radar; particles per mode 500,000, 250,000, 1,250,000, 250,000, 1,250,000 (order as in Figure 3d), 3,500,000 per replicate; BTO+BFO seeds 1, 2, 3, 4, 5, 6, 7, 8; code revision 1da8c7c; 88 threads; total runtime 48.0 min; peak memory 17,092 MiB. Hypotheses enabled: radar-fix (parameters in run.json). Sensitivity run 'next-i-repro-radar', varying replicates 4 to 8.

Reproduce with one command from the repository root: make hypothesis H=radar-fix

Plain: Core run I, family 'Davey dynamics'. In the early flight, the aircraft moves as in the Davey model, with Mach 0.73 to 0.84 throughout. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.891, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.

Technical: run next-i-repro-radar; stack: run D stack + speed rule (see the method page); binary 1da8c7c, x86_64-linux (deskstar); seeds 1-8 x 3,500,000 particles; ln Z -136.73; split-half mean over 35 balanced partitions 0.891 (range 0.853-0.927) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.

Parameters and model choices against Davey et al. (2016)

43 parameters and choices. 31 match the published model; 12 differ and are marked. A difference is not by itself a defect: some are deliberate and documented (ERA5 for the unavailable ACCESS-G, IGRF-14 for NOAA, the sampler), some are the subject of the run (a widened Mach range), and some are worth fixing. Page numbers refer to the book; the full table with citations and notes is written beside this report as parameters.csv.

	Parameter	This run	Davey et al. (2016)
	Dynamics		
	Mach OU reversion rate (1/s)	0.01058	0.01058
	Mach OU noise strength (1/s)	2.05e-07	2.05e-07
	Control-angle OU reversion rate (1/s)	0.009792	0.009792
	Control-angle OU noise strength (rad^2/s)	4.074e-08	4.074e-08
	Wind-error OU reversion rate (1/s)	0.001087	0.001087
	Wind-error OU noise strength (kn^2/s)	0.07021	0.07021
	Manoeuvre time constant, Jeffreys support (h)	[0.1, 10]	[0.1, 10]
	Mach after an acceleration, uniform on (Mach)	[0.73, 0.84]	[0.73, 0.84]
	Altitude after a vertical manoeuvre, uniform on (ft)	[25000, 43000]	[25000, 43000]
	Altitude step (ft)	1000	1000
	Bank angle in turns (deg)	15	15
	Mach rate during an acceleration (Mach/s)	0.00166667	0.00166667
	Vertical rate during a climb or descent (ft/s)	66.6667	66.6667
	Mean time to lateral-navigation reversion (s)	31162.2	31162.2
	Cruise integration step (s)	10	10
▶	Manoeuvre integration step (s)	5	1
	Prior		
	Prior epoch (UTC)	2014-03-07T18:01:49Z	2014-03-07T18:01:49Z
▶	Position standard deviation (NM)	0.5	0.4
	Track standard deviation (deg)	1	1
	BFO bias prior mean (Hz)	150	150
	BFO bias prior standard deviation (Hz)	25	25
▶	Initial Mach set point	-	[0.82, 0.03]
▶	Prior mean position and track	-	not tabulated
	Autopilot-mode prior weights	[[true heading, 1], [magnetic heading,…	equal over the five modes
	Measurement		
	BTO fixed offset (us)	495679	495679
	BTO noise sd, R1200 (us)	29 at 7 epochs	29
	BTO noise sd, anomalous R1200 (us)	43 at 2 epochs	43
▶	BTO noise sd, R600 (us)	63 at 1 epoch	62
	BFO noise sd (Hz)	7 at 11 epochs	7
	BFO bias treatment	constant per trajectory, Kalman-margina…	constant per trajectory, Kalman-marginalised
	Prior		
	BFO bias drift rate (Hz^2/s)	0	0
	Measurement		
	BTO bias treatment	-	fixed per-flight T_channel only; no bias st…
▶	Vertical rate in the BFO model	True	not modelled
	00:19 BFO	-	not used; believed unreliable
	Model choice		
▶	Weather source	config/../data/era5-wind-temperature.bin	ACCESS-G, 150-500 hPa (about 18,000-44,000…
▶	Magnetic declination source	config/../data/igrf14-declination.bin	NOAA
▶	Resampling scheme	one filter per autopilot mode combined…	randomised branching: weight >= eta branche…
	Manoeuvre-type dependence	-	turns, accelerations and vertical manoeuvre…
	Turn extent (deg)	-	uniform on -180 to +180
▶	Cost-index speed mode	-	implemented but unused in most experiments;…
▶	Fuel and endurance	{'factor_mean': 1.0004, 'factor_sd': 0.…	infinite fuel; constraints applied to the p…
	Sampler		
▶	Fuel proposal	endurance (deadline unix 1394237459, pr…	n/a (no fuel model)
	Model choice		
	Observation cutoff	m0019b	00:19 (BTO only at 00:19)

▶ marks a difference. Rows whose value is prose or a file identifier cannot be compared by string match; for those the reference file asserts the relationship explicitly and parameters.csv records why. Quantities held per epoch rather than per run — the BTO and BFO measurement standard deviations — are in data/satcom-observations.csv and are compared in the note column of the CSV.

Plain: Core run I, family 'Davey dynamics'. In the early flight, the aircraft moves as in the Davey model, with Mach 0.73 to 0.84 throughout. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.891, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.

Technical: run next-i-repro-radar; stack: run D stack + speed rule (see the method page); binary 1da8c7c, x86_64-linux (deskstar); seeds 1-8 x 3,500,000 particles; ln Z -136.73; split-half mean over 35 balanced partitions 0.891 (range 0.853-0.927) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.

Where was MH370 at 00:19 UTC?

A reproduction of the DSTG Bayesian aircraft-position pdf (Davey et al. 2016, Fig. 10.3)

Core run I, family 'descent and climb'. In the early flight, the aircraft can descend to between 2,000 and 10,000 ft and then climb again, before 18:18. Speeds are as for free flight. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight.

This report recreates the probability density function (pdf) of the aircraft's latitude at the final satellite handshake, 00:19 UTC on 8 March 2014, from the model published by Davey, Gordon, Holland, Rutten and Williams, Bayesian Methods in the Search for MH370 (Springer, 2016). The aircraft state is filtered from the penultimate radar point at 18:01:49 UTC through all Inmarsat burst timing (BTO) and burst frequency (BFO) measurements, using the published cruise, manoeuvre and measurement models.

With BTO and BFO the recreated pdf has median 37.06°S and mode 37.35°S, with 95% of probability between 31.06°S and 38.34°S. The published curve has median 37.53°S, mode 37.90°S and 95% range 34.90–39.38°S. The two densities overlap by 76%. This curve is centred 0.5° further north and carries 32% of probability in the published northern shoulder between 34.5°S and 36.5°S (25% in the book).

Relative to the same family in run D, the median moves 0.11° north, the 34.5–36.5°S shoulder holds 32% of probability (base 20%), and the two curves overlap by 79%.

The curve pools 8 independent replicates of 3,500,000 particles. Single replicates overlap each other by 54%–81% and their medians span 1.63°. Over all balanced partitions of the 8 replicates the split-half overlap averages only 79.4% (range 72.5% to 85.5%) against the 8-replicate floor of 0.924, so the pooled curve is NOT converged at this particle count and quantities read off it carry that caveat.

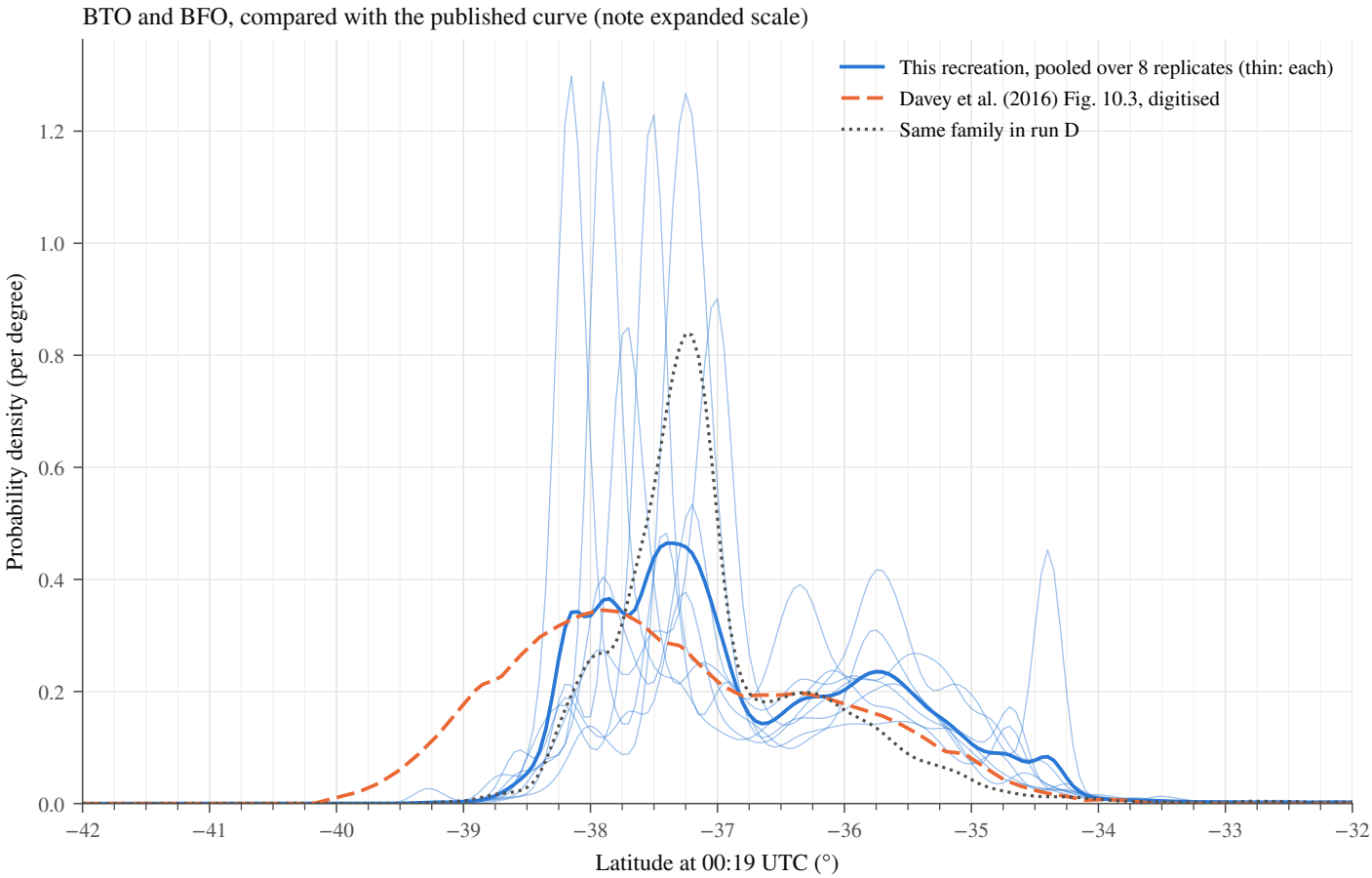


Figure 1. Latitude pdf of the aircraft at 00:19:37 UTC, the form of Davey et al. Fig. 10.3. BTO and BFO, with the published curve. Densities are weighted particle histograms smoothed with a 0.1° Gaussian; each particle carries its own altitude. Evidence: BTO at every epoch except the 18:39 and 23:15 C-channel calls; BFO (bottom only) at 18:28, 18:39, 19:41–22:41, 23:15 and 00:11. This run also uses: primary-radar positions and track segments 18:04-18:22 scored as measurements; fuel model internal-v1, two tanks, one-engine flight after the first flame-out (hold-taper drift-down), paths dry before 00:11 rejected. No debris, drift, search or end-of-flight evidence is used. Run conditions as stated at the top of this page.

Plain: Core run I, family 'descent and climb'. In the early flight, the aircraft can descend to between 2,000 and 10,000 ft and then climb again, before 18:18. Speeds are as for free flight. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.794, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.

Technical: run next-i-descent-climb; stack: run D stack + speed rule (see the method page); binary 8d7cf7a, x86_64-linux (deskstar); seeds 1-8 x 3,500,000 particles; ln Z -137.46; split-half mean over 35 balanced partitions 0.794 (range 0.725-0.855) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.

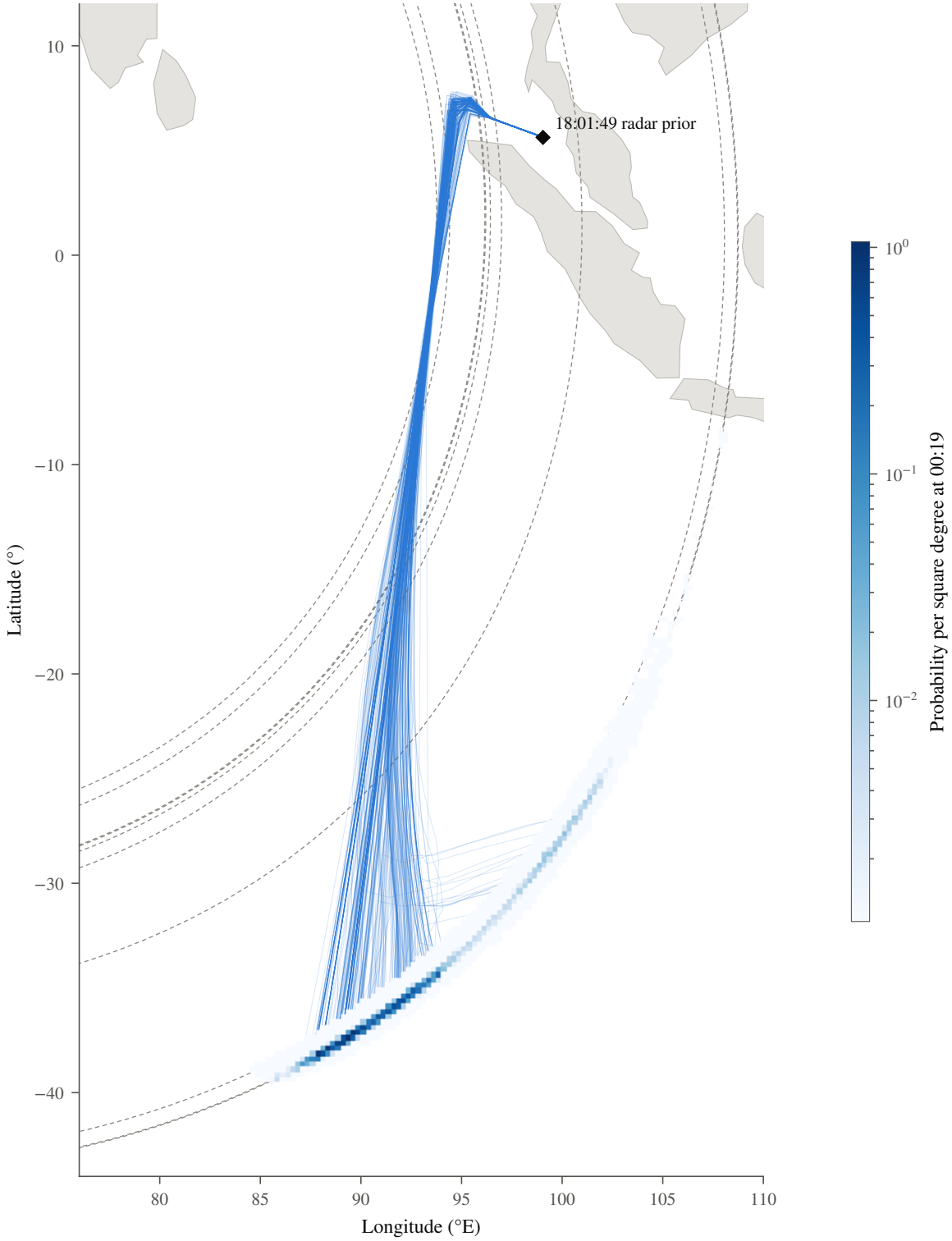
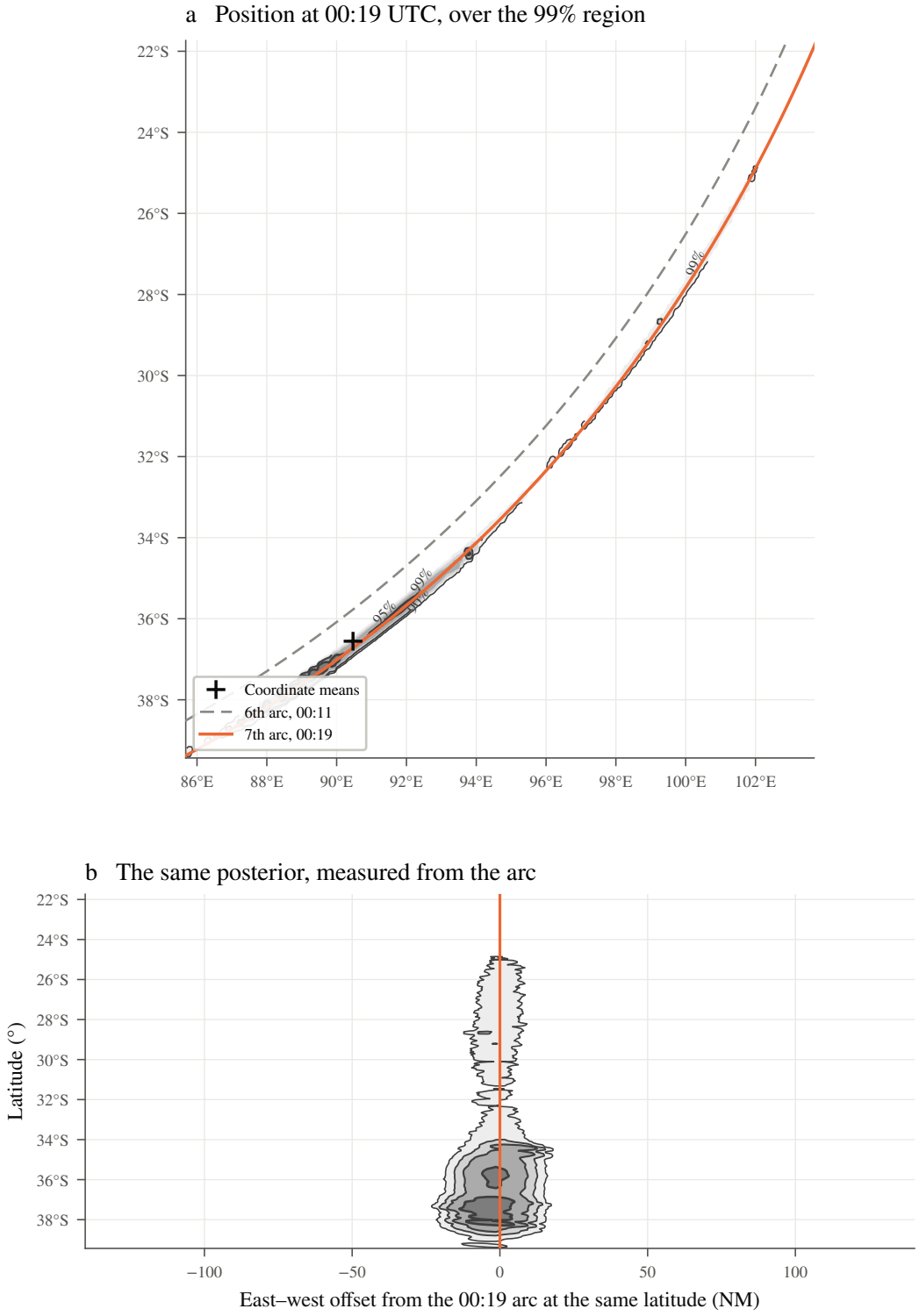


Figure 2. Posterior aircraft position at 00:19 UTC with BTO and BFO, pooled over 8 replicates (blue shading, logarithmic), and 400 routes drawn in proportion to posterior weight from replicate seed 1 (thin lines, 10-minute vertices). Dashed grey curves are the BTO arcs at a reference altitude of 35,000 ft; they are drawn for orientation only. Land: Natural Earth 1:110m.

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DIFFERENCES FROM DAVEY

Mach after an acceleration, uniform on: [0.7, 0.87]

Manoeuvre integration step: 5

Position sd: 0.5

Initial Mach set point: -

Prior mean position and track:

BTO noise sd, R600: 63 at 1 epoch

Vertical rate in the BFO model: True

Weather source: era5-wind-temperature.bin

Magnetic declination source: igrf14-declination.bin

Resampling scheme: one filter per autopilot …

Cost-index speed mode:

Fuel and endurance: {'factor_mean': 1.0004, '…

Fuel proposal: endurance (deadline unix …

book values: see the parameter page

POSTERIOR AT 00:19

median 37.06°S, mode 37.35°S

95% 31.06–38.34°S

replicates 8, split-half 0.743

largest single draw 0.00% of weight

99% region 18,114 NM²

95% region 8,523 NM²

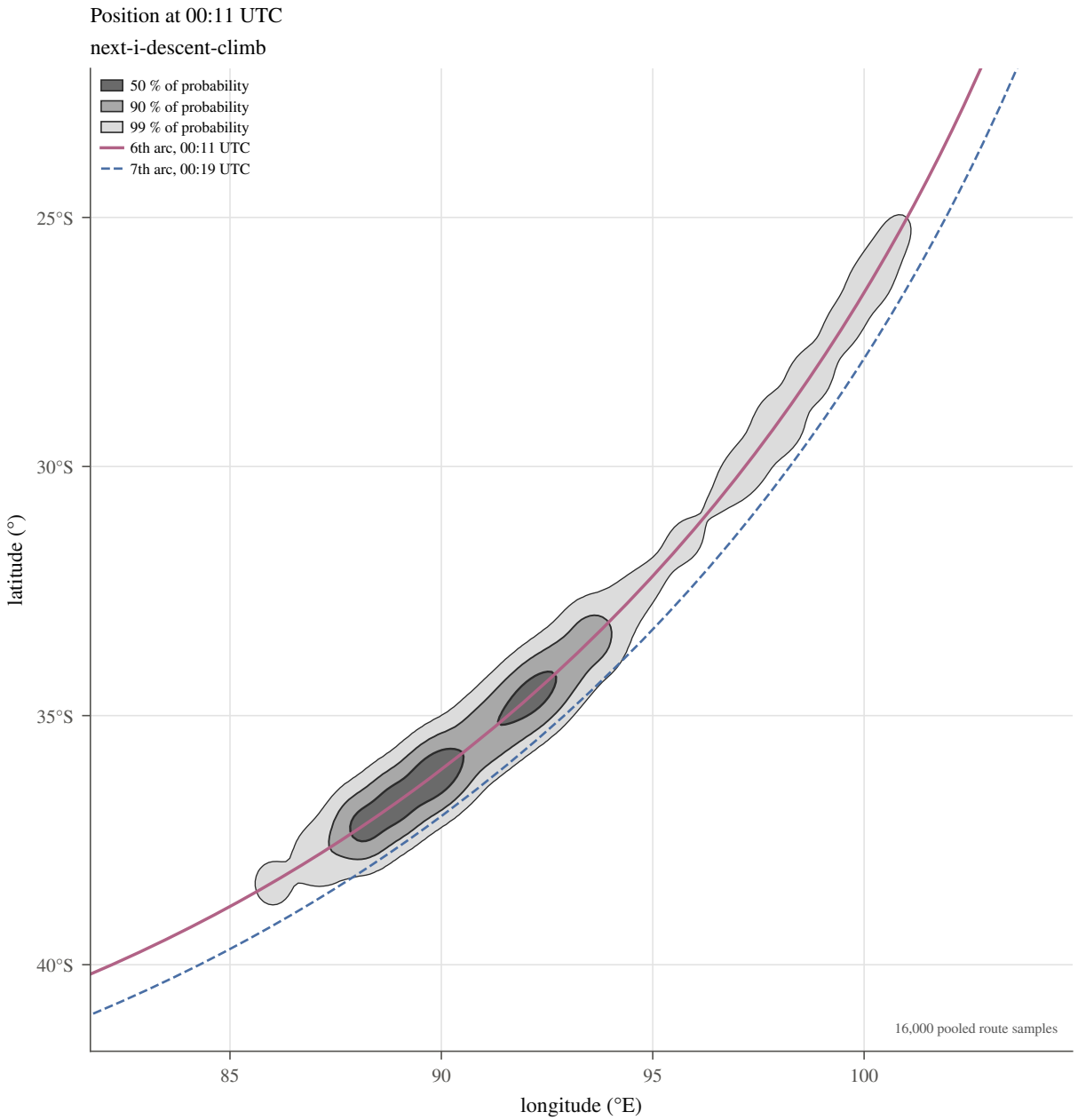
90% region 6,028 NM²

50% region 1,612 NM²

Figure 3. Joint latitude-longitude posterior at 00:19 UTC, drawn as highest-posterior-density regions: the 50% curve bounds the smallest area holding 50% of the probability, and so on outward, so a region here is the smallest place to look for that confidence rather than an interval in either coordinate alone. Binned at 0.05° and smoothed over three bins. Panel a spans the bounding box of the 99% region, so sparse outlying cells are excluded; enclosed areas (NM², at the view's mean latitude) are 99% 18,114, 95% 8,523, 90% 6,028, 50% 1,612. Panel b replots the same particles against their east-west offset from the 00:19 arc, because the region is about 0.3° wide and 18° long and the inner contours are not separable at a true aspect ratio. Arcs are the reference BTO loci at 35,000 ft; 00:19 is the 7th. Land: Natural Earth 1:110m.

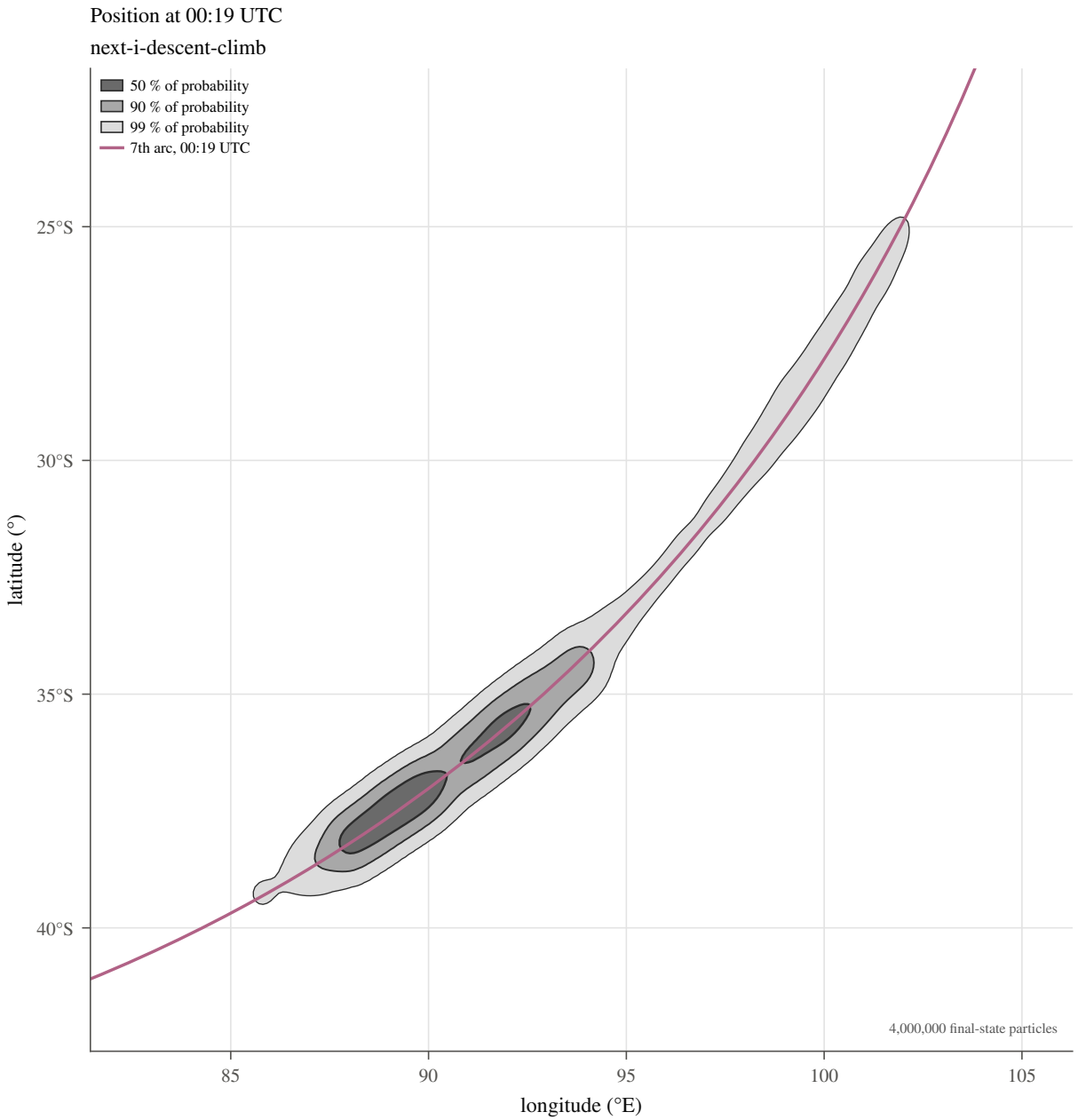
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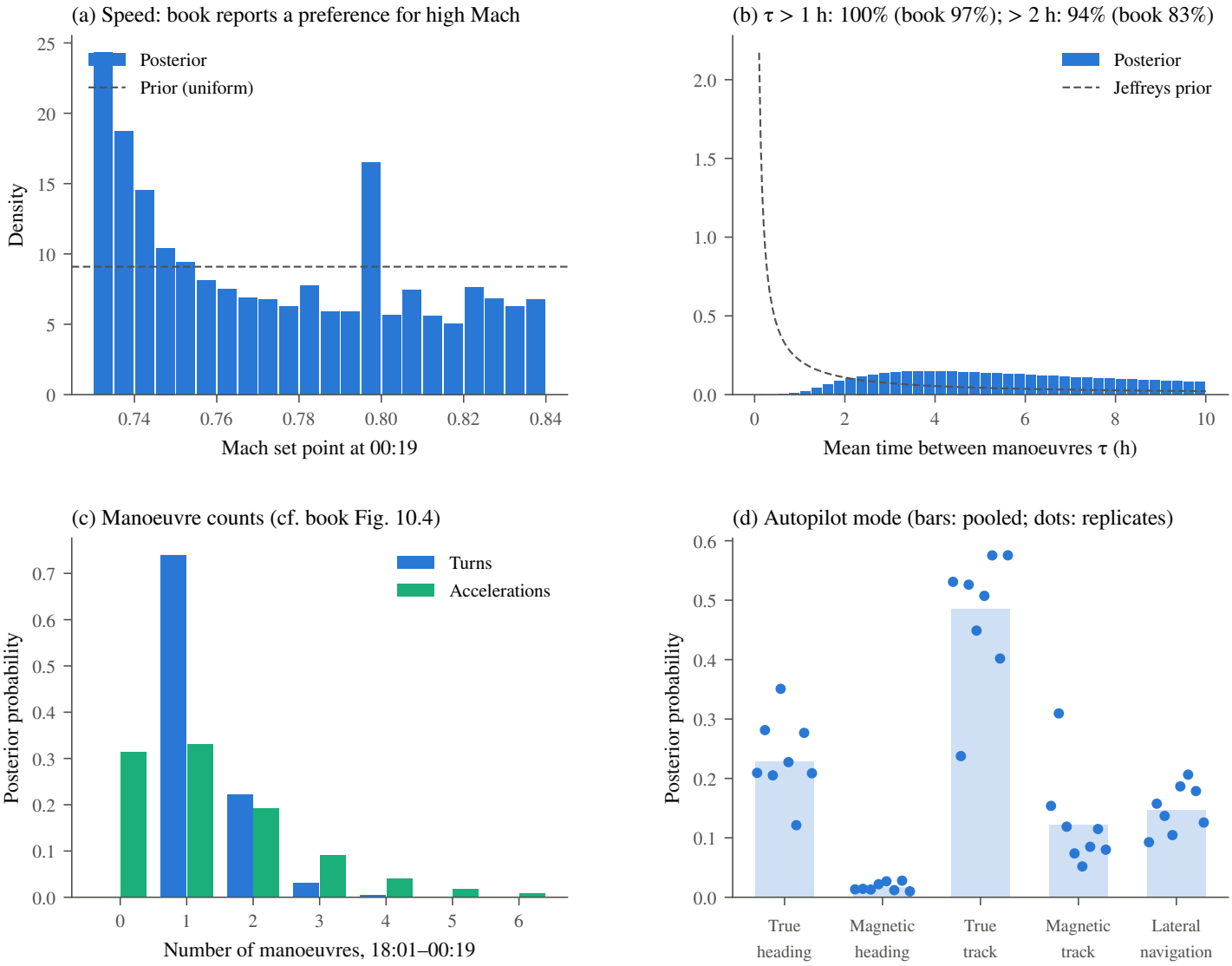
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Posterior behaviour compared with the book



	This recreation	Davey et al. (2016)
Median latitude	37.06°S	37.53°S
Mode	37.35°S	37.90°S
95% interval	31.06–38.34°S	34.90–39.38°S
95% width	7.28°	4.47°
Overlap with published pdf	76% (replicates 55%–72%)	—
Median Mach at 00:19	0.770	upper part of 0.73–0.84

Figure 3 and Table 1. Posterior summaries with BTO and BFO, pooled over replicates, alongside the statements in Davey et al. (2016), sec. 10.2–10.4. Book values of the latitude pdf come from the digitised Fig. 10.3 (bottom); other book values are quoted from the text.

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Method, convergence and reproduction

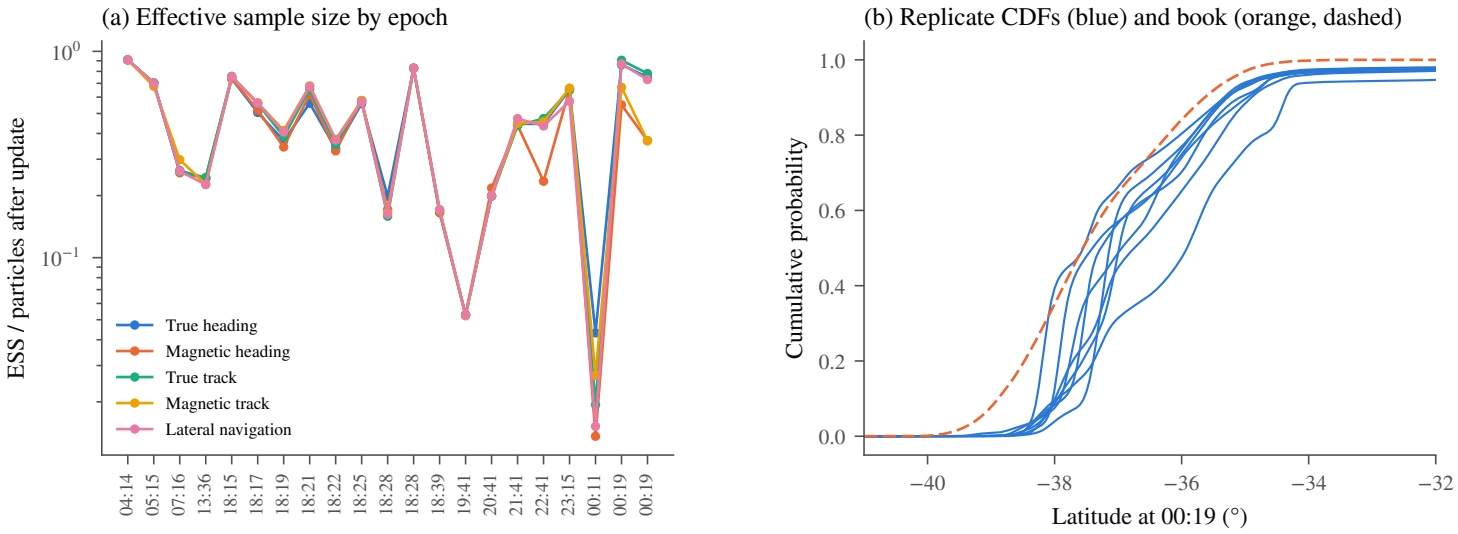


Figure 4. (a) Effective sample size after each measurement update, averaged over replicates, per autopilot-mode filter. The narrowest bottlenecks are the 18:39 C-channel BFO (the southward turn) and the 19:41 BTO arc. (b) Latitude CDFs of the 8 independent replicates; the largest difference between any two is 0.402 (Kolmogorov distance).

Model. The state, dynamics and likelihood follow Davey et al. (2016) chapters 4–8: prior at 18:01:49 UTC with 0.5 NM position and 1° track uncertainty, Mach uniform on 0.73–0.84, altitude on 1,000 ft levels from 25,000 to 43,000 ft; five autopilot modes (true/magnetic heading, true/magnetic track, lateral navigation with a one-off reversion to heading hold); Ornstein–Uhlenbeck fluctuations of Mach, control angle and wind error with the published rates; turns, speed changes and climbs arriving with exponential gaps of common mean τ (Jeffreys prior on 0.1–10 h) at 15° bank, 0.1 Mach per minute and 4,000 ft per minute. BTO is Gaussian with the tabulated per-message errors (29, 43 or 63 µs); BFO has 7 Hz noise and an unknown constant bias (prior 150 ± 25 Hz) marginalised exactly per particle with a Kalman filter.

Substitutions. The book's ACCESS-G weather is replaced by ERA5 winds and temperatures; magnetic declination is IGRF-14. The prior mean position and track are reconstructed from the book's radar figures because they are not tabulated. Davey et al. used variable-rate branching; this recreation uses fixed-size particle filters with systematic resampling when ESS falls below 50% of the particles. The book warns that conventional resampling collapses static parameters (sec. 10.4). Both are therefore handled exactly: one filter per autopilot mode, combined in proportion to its estimated marginal likelihood, and a Gibbs refresh of τ from its path conditional after every resampling step. Without them, a plain bootstrap filter of one million particles retained only about one hundred distinct prior draws, and an earlier recreation in this project found tails and turn counts that changed with the random seed.

Run. Configuration next-i-descent-climb; particles per mode 500,000, 250,000, 1,250,000, 250,000, 1,250,000 (order as in Figure 3d), 3,500,000 per replicate; BTO+BFO seeds 1, 2, 3, 4, 5, 6, 7, 8; code revision 8d7cf7a; 88 threads; total runtime 55.4 min; peak memory 17,863 MiB. Hypotheses enabled: radar-fix (parameters in run.json). Sensitivity run 'next-i-descent-climb', varying cas_envelope_kt (not declared) to [210.0, 330.0]; mach_range [0.73, 0.84] to [0.7, 0.87]; replicates 4 to 8.

Reproduce with one command from the repository root: make hypothesis H=radar-fix

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43 parameters and choices. 30 match the published model; 13 differ and are marked. A difference is not by itself a defect: some are deliberate and documented (ERA5 for the unavailable ACCESS-G, IGRF-14 for NOAA, the sampler), some are the subject of the run (a widened Mach range), and some are worth fixing. Page numbers refer to the book; the full table with citations and notes is written beside this report as parameters.csv.

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	Wind-error OU reversion rate (1/s)	0.001087	0.001087
	Wind-error OU noise strength (kn^2/s)	0.07021	0.07021
	Manoeuvre time constant, Jeffreys support (h)	[0.1, 10]	[0.1, 10]
▶	Mach after an acceleration, uniform on (Mach)	[0.7, 0.87]	[0.73, 0.84]
	Altitude after a vertical manoeuvre, uniform on (ft)	[25000, 43000]	[25000, 43000]
	Altitude step (ft)	1000	1000
	Bank angle in turns (deg)	15	15
	Mach rate during an acceleration (Mach/s)	0.00166667	0.00166667
	Vertical rate during a climb or descent (ft/s)	66.6667	66.6667
	Mean time to lateral-navigation reversion (s)	31162.2	31162.2
	Cruise integration step (s)	10	10
▶	Manoeuvre integration step (s)	5	1
	Prior		
	Prior epoch (UTC)	2014-03-07T18:01:49Z	2014-03-07T18:01:49Z
▶	Position standard deviation (NM)	0.5	0.4
	Track standard deviation (deg)	1	1
	BFO bias prior mean (Hz)	150	150
	BFO bias prior standard deviation (Hz)	25	25
▶	Initial Mach set point	-	[0.82, 0.03]
▶	Prior mean position and track	-	not tabulated
	Autopilot-mode prior weights	[[true heading, 1], [magnetic heading,…	equal over the five modes
	Measurement		
	BTO fixed offset (us)	495679	495679
	BTO noise sd, R1200 (us)	29 at 7 epochs	29
	BTO noise sd, anomalous R1200 (us)	43 at 2 epochs	43
▶	BTO noise sd, R600 (us)	63 at 1 epoch	62
	BFO noise sd (Hz)	7 at 11 epochs	7
	BFO bias treatment	constant per trajectory, Kalman-margina…	constant per trajectory, Kalman-marginalised
	Prior		
	BFO bias drift rate (Hz^2/s)	0	0
	Measurement		
	BTO bias treatment	-	fixed per-flight T_channel only; no bias st…
▶	Vertical rate in the BFO model	True	not modelled
	00:19 BFO	-	not used; believed unreliable
	Model choice		
▶	Weather source	config/../data/era5-wind-temperature.bin	ACCESS-G, 150-500 hPa (about 18,000-44,000…
▶	Magnetic declination source	config/../data/igrf14-declination.bin	NOAA
▶	Resampling scheme	one filter per autopilot mode combined…	randomised branching: weight >= eta branche…
	Manoeuvre-type dependence	-	turns, accelerations and vertical manoeuvre…
	Turn extent (deg)	-	uniform on -180 to +180
▶	Cost-index speed mode	-	implemented but unused in most experiments;…
▶	Fuel and endurance	{'factor_mean': 1.0004, 'factor_sd': 0.…	infinite fuel; constraints applied to the p…
	Sampler		
▶	Fuel proposal	endurance (deadline unix 1394237459, pr…	n/a (no fuel model)
	Model choice		
	Observation cutoff	m0019b	00:19 (BTO only at 00:19)

▶ marks a difference. Rows whose value is prose or a file identifier cannot be compared by string match; for those the reference file asserts the relationship explicitly and parameters.csv records why. Quantities held per epoch rather than per run — the BTO and BFO measurement standard deviations — are in data/satcom-observations.csv and are compared in the note column of the CSV.

Plain: Core run I, family 'descent and climb'. In the early flight, the aircraft can descend to between 2,000 and 10,000 ft and then climb again, before 18:18. Speeds are as for free flight. After 18:40 the model is Davey's, with the fuel model, radar and one-engine flight. The estimate is not converged (split-half 0.794, below the floor of 0.924 for 8 runs). The physics of the one-engine phase is provisional. The latitude at 00:19 uses both BTO values and no BFO.

Technical: run next-i-descent-climb; stack: run D stack + speed rule (see the method page); binary 8d7cf7a, x86_64-linux (deskstar); seeds 1-8 x 3,500,000 particles; ln Z -137.46; split-half mean over 35 balanced partitions 0.794 (range 0.725-0.855) against the 8-replicate floor 0.924; 00:19 treatment: core, Davey both-BTO no-BFO; one-engine dynamics s7 + s8 hold-then-taper (PROVISIONAL); overlay: the same family in run D (4 seeds; for waypoints, run D's 48 airway routes); report built 12 Oct 2026.